



IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

PRI12NP - Integrated Nuclear Engineering Project 02 - Simulation of an Experimental Setup

Under the guidance of :
Professor(s) Amanda PORTA, Mariangela SETTIMO

Radiation-Matter Interaction TP GEANT4

ABSTRACT:

Gamma (γ) particle and electron (e^-) particle radiation is hereby simulated to analyze their energy deposition effects in a liquid Xenon (Xe) tracker.

It was found that the use of lead (Pb) shielding as thick as 1cm may be partially effective in preventing the deposition of radiation energy in the Xenon tracker.

The use of Monte Carlo simulation software GEANT4, and of statistical analysis ROOT, both by the CERN, was needed to estimate the energy deposition to number of events spectra.

Date	Peer 1	Peer 2	Peer 3
17 Apr 2026	Javier Alonzo ROMO LEON		

The present document formatting has been aligned to meet standard practices as per ISO9001 in support of future standardization processes.

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1. Introduction.

The present document compiles class exercises reviewed in the context of Master of Science cohorts on Integrated Nuclear Engineering Project: 02 - Simulation of an Experimental Setup as taught by Professor(s) Amanda PORTA, Mariangela SETTIMO, with the intent of:

- Documenting their resolution methodology,
- Document lessons learned,
- Document relevant bibliography,
- Provide a quick entry point to solve related scenarios in future professional endeavors.

1.1. Problem summary and location of solutions.

The activity as instructed in section 3 required making use of GEANT4 and ROOT, both provided by CERN as open source software, to simulate the energy deposition of radiation particles in a particle tracker system.

This is to simulate a tridimensional cylindrical volume considered to be of liquid Xenon, from hereon referred to as the particle “tracker” substance. E.g. as found in Xenon medical imaging systems. Its properties are shown in Figure 1. Its geometry is defined among the source code as “trackerLength” and “trackerRadius”.

The aforementioned may or may not feature a protective lid of lead (Pb), which is to be referred to as the “target”, shown in Figure 4. Its geometry is defined among the source code as “targetLength” and “targetRadius”.

This system is to be exposed to a beam of particles in configurations further specified in more detail in section 3, and demonstrated in section 4 using Monte Carlo simulations in GEANT4. This is shown in Figure 1 and Figure 2.

```
// Liquid xenon defined using class G4Material
fTrackerMaterial =
new G4Material("XenonLiquid", z=54, a=131.29*g/mole, density= 3.06*g/cm3,
               kStateLiquid, temperature= 293.15*kelvin, pressure= 1*atmosphere);
```

Figure 1: Liquid Xenon properties.

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Date: 17 April 2026

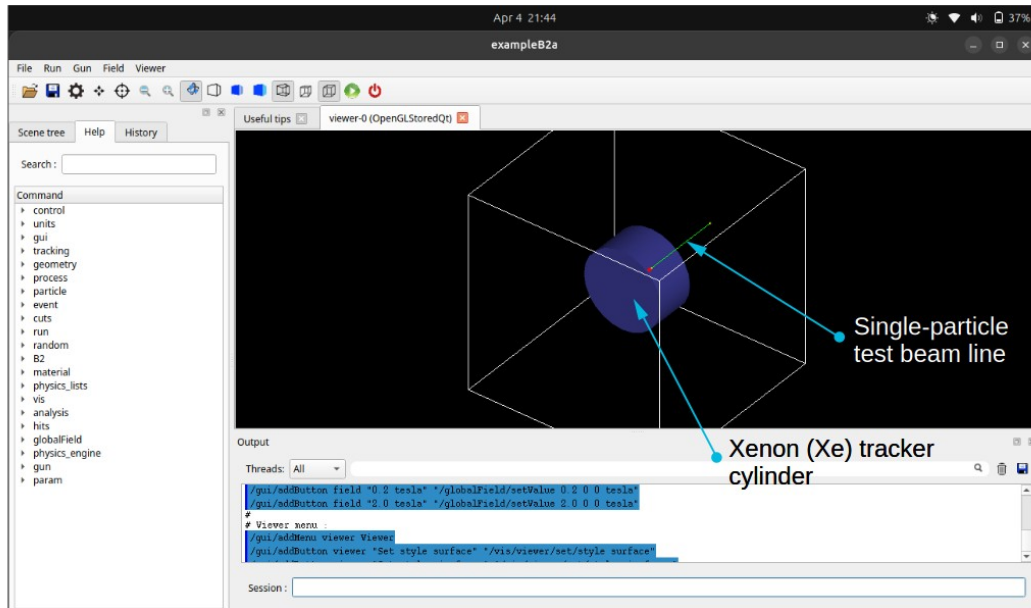


Figure 2: Isometric view of the particle tracker system.

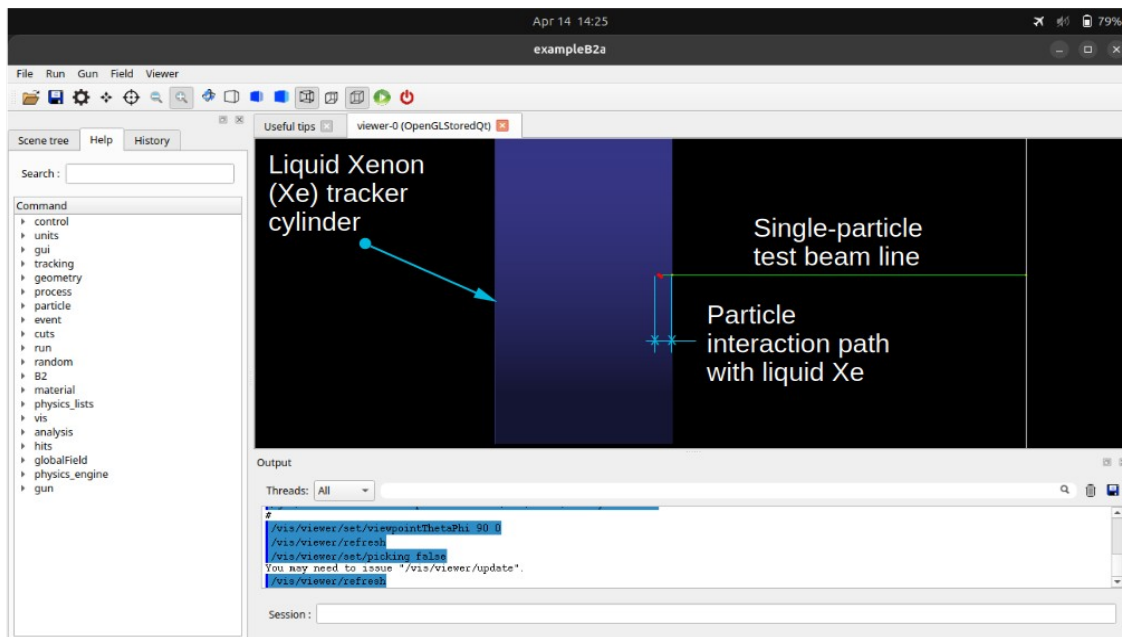


Figure 3: Side view of the particle tracker system, no Pb lid.

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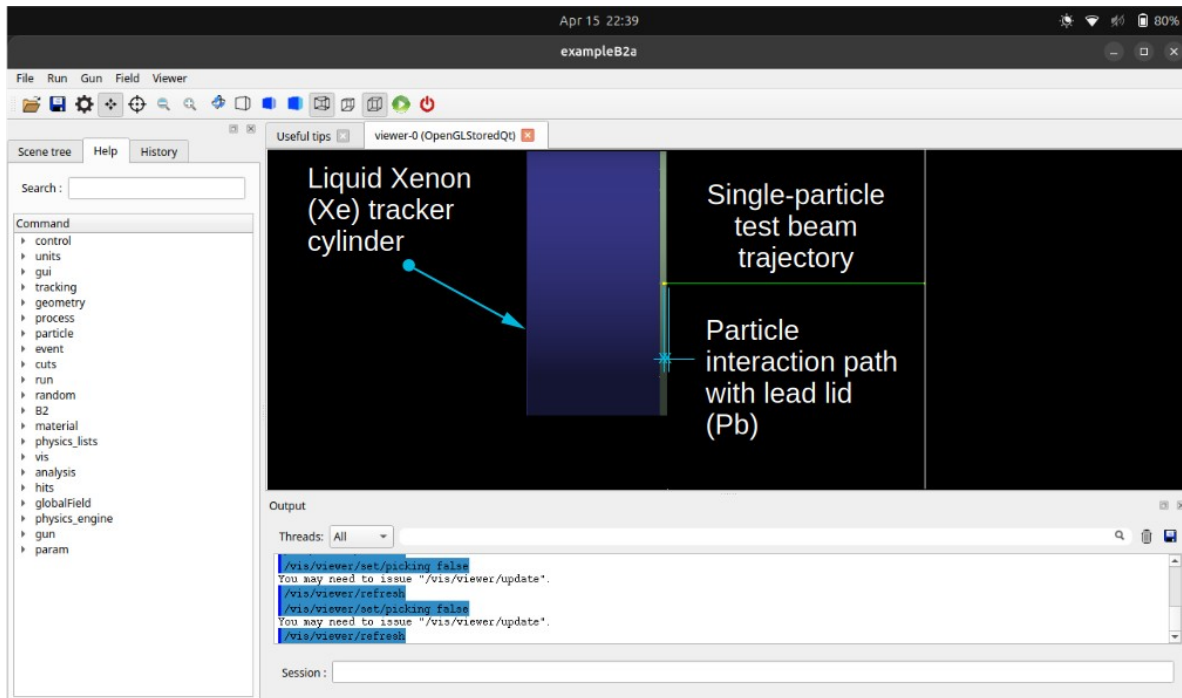
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1.2. Repositories and communications.

The GEANT4 and ROOT source code needed to execute this project is presently backed up and upkept in the following repository:

<https://github.com/J4v1987/PRI12NP---Integrated-Nuclear-Engineering-Project/tree/main/TP%20GEANT4/B2a>

Professionals in the nuclear engineering realm and interested audience are invited to participate in the development of this script or in commenting and/or correcting errata. Do consider contacting the repository maintainer via dedicated communications found in:

<https://sites.google.com/view/b-eng-jarl/home>

2. Key Definitions and Abbreviations

Term or Abbreviation	Definition
Candidate	Referring to the author of the present report: Javier Alonzo ROMO LEON, also entrusted to perform the Laboratory Specifications provided in section 3, this in the context of Master of Science cohorts on Nuclear Engineering hosted by the Institution.
CERN	Conseil Européen pour la Recherche Nucléaire (The European Council for the Nuclear Research). More details at: https://home.cern/about/who-we-are/our-history
e^-	Symbol referring to an electron particle, or merely “electron(s)”.
GEANT4	A toolkit to create simulations of the passage of particles or radiation through matter. Applications built on GEANT4 can simulate any setup or detector and radiation source, and record chosen output of physical quantities due to source particles and secondaries interacting with the material of the setup.
Institution	Refers to “L’Institut Mines-Telecom”, branded as “IMT Atlantique” for the specific case of its campus at Bretagne-Pays de la Loire, France, whom are a Federal Institute of Technological Universities of science and management for sustainable economic development: https://www.imt.fr/en/imt/our-identity/
ROOT	Terminal-based software for the processing and scientific analysis of large amounts of data.

3. Laboratory Specifications

Radiation-Matter Interaction – Master M1 NE TP GEANT4

A. Porta, M. Settimo (April 2026)

Installation of GEANT4 application

The GEANT4 documentation is available on the CERN web <https://geant4.web.cern.ch/>. GEANT4 is already installed on a common disk readable for all users having a computer account at IMT Atlantique.

- In this TP, you will create your own application by copying the GEANT4 example from the moodle in your *geant4* directory:

Create your *geant4* directory: `mkdir geant4`

Then: `cd geant4`

Download the GEANT4 code from

<https://moodle.imt-atlantique.fr/course/view.php?id=2554>

decompress the file and put the obtained *B2a* directory in your *geant4* directory (as you did for the C++ exercise)

- To define the GEANT4 environment in your *geant4* directory write SETUP GEANT4:
You will have to type this command every time you open a new shell terminal.
- To build your application, go to your *B2a* directory and then create a new directory:
`mkdir B2a-build`
- Go to this new directory: `cd B2a-build`
This new directory will be used to build your application i.e. to compile your program and create an executable file.
- Generate the Makefiles needed to build the *B2a* application:
`cmake ../`
- To compile your application and create the executable file, you should type in the *B2a-build* directory: `make`
- You can now run your application in the *B2a-build* directory: `./exampleB2a`

Whenever you need to modify the source code, you have to recompile by typing *make* in the *B2a-build* working directory.

GEANT4 Simulation

Take time to carefully analyze the output text of the execution of your application, it will help you to understand how the "tracking" is working in GEANT4.

Figure 5: Page 1 from Laboratory Specifications.

- When you launch *exampleB2a*, the macro *vis.mac* is automatically called by your GEANT4 application. Open the file which is in the current *B2a-build* working directory by typing: *kate vis.mac &* and try to understand the list of command lines written in this file.
- You can run all these command lines directly in interactive session. Try, for example to type: */vis/viewer/set/viewpointThetaPhi 50 120 deg* after the prompt *Idle*. The viewing angle of the device must be changed.
- To visualize the histograms showing the results; in your *xterm* launch *root simulation.root*, then type *new TBrowser*, then click on *ROOT Files*, then click on histogram *jH1*. The ROOT software documentation is available at the following address <http://root.cern.ch/>.

The objective of this TP is to understand the GEANT4 software as Monte Carlo simulation of the radiation-matter interaction processes. For this, an experimental device consisting of a lead target (*Target*) and a liquid xenon detector (*tracker*) inside a volume of air (*World or Mother volume*) has already been implemented. To know the characteristics of this device, look at the classes defined in the *src* directory. You will have to modify the source code to define the simulation configuration and parameters (geometry, filling of histograms ...)

Simulation of 1 MeV γ -rays

1. Plot the histogram of energy deposition in liquid xenon detector without lead. To do this, type e.g. */run/beamOn 100000* at the command prompt *Idle>* to generate 100 000 gamma of 1 MeV. Repeat the simulation for different thicknesses of the volume of liquid xenon (1 cm, 10 cm, ...). The thickness of xenon is already defined in the class *B2aDetectorConstruction.cc* at the following code line:

```
trackerLength = 20*cm;
```

What do you observe?

Important remark: to accelerate the simulation of a large number of events $\sim 10^5$, it is better to remove the visualization interface and the text output. To do this, you can create your own *run.mac* file with only the following parameters inside:

```
/run/initialize  
  
# Define level of verbosity  
/tracking/verbose 0  
  
# Define particle type  
/gun/particle gamma  
# Define particle energy  
/gun/energy 1 MeV  
# start run with one particle generated  
/run/beamOn 100000
```

To use this *run.mac* file to initialize the simulation parameters instead of the default *vis.mac* file you should launch: *./exampleB2a run.mac*

Figure 6: Page 2 from Laboratory Specifications.

2. Add the lead target already defined in *B2aDetectorConstruction.cc*, to do this uncomment the following lines of code:

```
// new G4PVPlacement(0,           // no rotation
//                    positionTarget, // at (x,y,z)
//                    fLogicTarget,  // its logical volume
//                    "Target",      // its name
//                    worldLV,       // its mother volume
//                    false,         // no boolean operations
//                    0,             // copy number
//                    fCheckOverlaps); // checking overlaps
```

Plot the histogram of energy deposition in liquid xenon. Is the effect of the target in agreement with what you can predict from radiation-matter calculation?

Simulation of 2 MeV electrons

Simulate 2 MeV electrons with and without lead target. Plot the histogram of energy deposition in liquid xenon in both cases. Comment on the effect of the lead target on the number of detected electrons and the shape of the energy spectrum.

Figure 7: Page 3 from Laboratory Specifications.

4. Experiment Results

4.1. Simulation of 1 MeV γ -rays, No Lead (Pb) Target

[Activity 1] A typical tracker – beam system as shown in Figure 2 was implemented, with:

trackerRadius = 20 [cm]

trackerLength = [1, 5, 10, 15, 20] [cm]

[Activity 2] 100,000 interactions were simulated per trackerLength instance, 500,000 data points in total, resulting in the chart seen in Figure 8 with tolerances as provided in Figure 9 relating Energy from 1 MeV γ -rays, E , deposited in the tracker, as a function of the tracker ‘thickness’, x , also known as the trackerLength:

$$E = -2.31 x^2 + 91.21 x + 8.35.$$

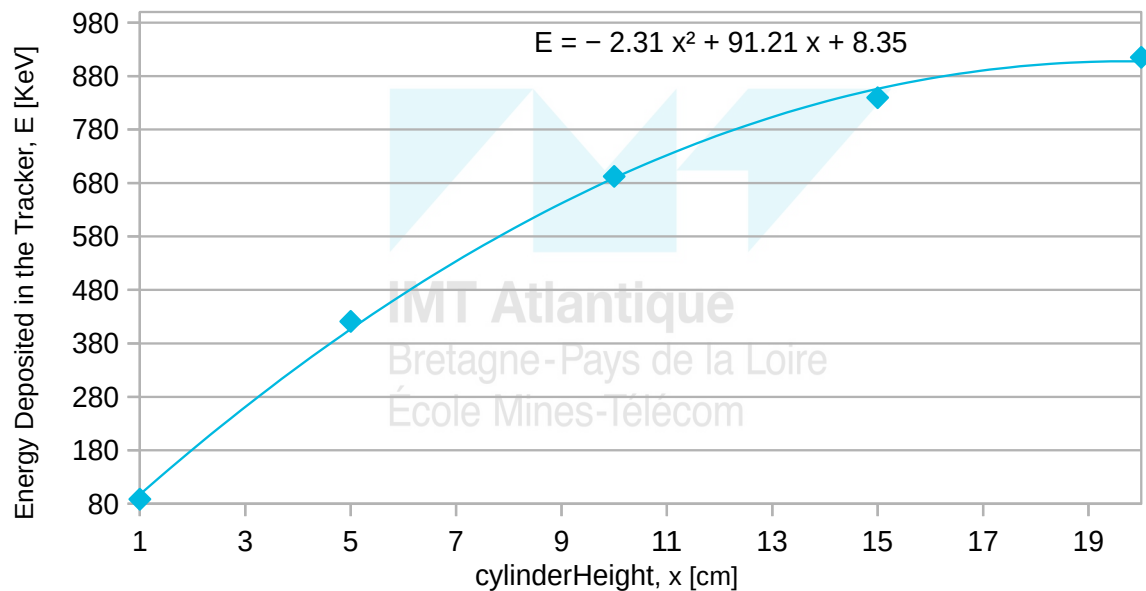


Figure 8: Energy deposited in Xe, 1 MeV γ - rays, no Pb lid.

Xe Tracker Radius [cm]	Tracker Cylinder height [cm]	Energy deposited [keV] (Population mean, μ)	$\pm\sigma$	Population
20.00	1.00	88.40	242.40	100,000.00
20.00	5.00	421.20	446.30	100,000.00
20.00	10.00	692.40	418.80	100,000.00
20.00	15.00	839.50	329.40	100,000.00
20.00	20.00	915.20	243.50	100,000.00

Deposited Energy, E_a [MeV]	Deposited Energy, E_a [J]	Xe Density, ρ [g/cm ³]	Xe Volume [cm ³]	Xe Mass [kg]	Xe Dose [Gy]
8,833,565.72	1.42E-06	3.06	1,256.64	3.85	3.68E-07
42,122,499.03	6.75E-06	3.06	6,283.19	19.23	3.51E-07
69,239,514.15	1.11E-05	3.06	12,566.37	38.45	2.88E-07
83,948,014.04	1.34E-05	3.06	18,849.56	57.68	2.33E-07
91,516,950.05	1.47E-05	3.06	25,132.74	76.91	1.91E-07

Figure 9: Energy deposited in Xe, 1 MeV γ - rays, no Pb lid.[1][2][3]

[Activity 3] Histograms were plotted to relate different bins of energy levels to the number of energy deposition events. The bins can be as many as the sqrt(population) or at least 10 bins, refer to the project script csv2root.C, variable: nbins. This activity was performed for all five (5) instances listed in Figure 9 and each corresponding histogram is presented in Appendix A.

4.2. Simulation of 1 MeV γ -rays, with Lead (Pb) Target

[Activity 4] A typical tracker (Xe) – target (Pb) – beam system as shown in Figure 4 was implemented, with:

targetRadius = 20 [cm]

targetLength = [0.2, 0.4, 0.6, 0.8, 1.0] [cm]

trackerRadius = 20 [cm]

trackerLength = 20 [cm]

[Activity 5] 100,000 iterations were simulated per targetLength instance, 500,000 data points in total, resulting in the chart seen in Figure 10 with tolerances as provided in Figure 11 relating Energy from 1 MeV γ -rays, E , deposited in the tracker, as a function of the target 'thickness', t :

$$E = 76.07 t^2 + 478.49 t + 919.04.$$

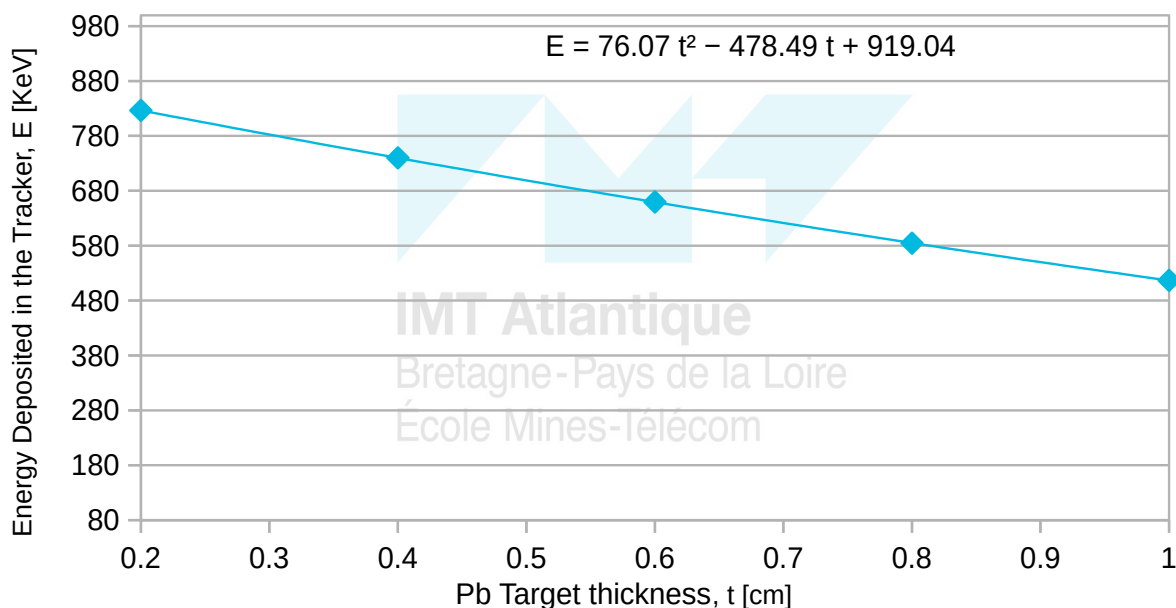


Figure 10: Energy deposited in Xe, 1 MeV γ - rays, with Pb target.

Xe & Pb Target Radius [cm]	Pb Target thickness [cm]	Energy deposited [keV] (Population mean, μ)	$\pm\sigma$	Population
20.00	0.20	826.30	341.90	100,000.00
20.00	0.40	739.90	402.20	100,000.00
20.00	0.60	659.60	438.40	100,000.00
20.00	0.80	584.50	458.80	100,000.00
20.00	1.00	516.80	467.10	100,000.00

Deposited Energy, E_d [MeV]	Deposited Energy, E_d [J]	Xe Density, ρ [g/cm ³]	Xe Volume [cm ³]	Xe Mass [kg]	Xe Dose [Gy]
82,627,099.99	1.32E-05	3.06	2.51	0.01	1.72E-03
73,986,265.52	1.19E-05	3.06	10.05	0.03	3.85E-04
65,958,406.32	1.06E-05	3.06	22.62	0.07	1.53E-04
58,447,622.77	9.36E-06	3.06	40.21	0.12	7.61E-05
51,682,806.61	8.28E-06	3.06	62.83	0.19	4.31E-05

Figure 11: Energy deposited in Xe, 1 MeV γ - rays, with Pb target.[1][2][3]

[Activity 6] Histograms were plotted to relate different bins of energy levels to the number of energy deposition events. The bins can be as many as the sqrt(population) or at least 10 bins, refer to the project script csv2root.C, variable: nbins. This activity was performed for all five (5) instances listed in Figure 11 and each corresponding histogram is presented in Appendix B.

4.3. Simulation of 2 MeV e^- -rays, No Lead (Pb) Target

[Activity 7] A typical tracker – beam system as shown in Figure 2 was implemented, with:
trackerRadius = 20 [cm]
trackerLength = [1, 5, 10, 15, 20] [cm]

[Activity 8] 100,000 iterations were simulated per trackerLength instance, 500,000 data points in total, resulting in the chart seen in Figure 12 with tolerances as provided in Figure 13 relating Energy from a 2 MeV e^- beam, E , deposited in the tracker, as a function of the tracker thickness, x , also known as the trackerLength:

$$E = -0.17 x^2 + 6.54 x + 1465.60.$$

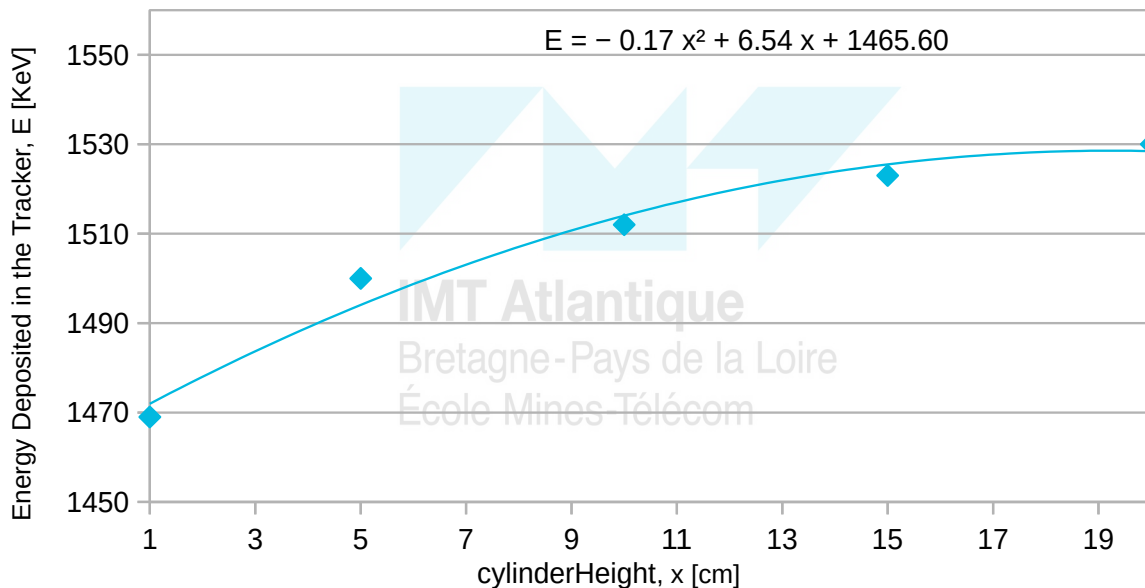


Figure 12: Energy deposited in Xe, 2 MeV e^- - beam, no Pb lid.

Xe Tracker Radius [cm]	Tracker thickness CylinderHeight [cm]	Energy deposited [keV] (Population mean, μ)	$\pm\sigma$	Population
20.00	1.00	1,469.00	596.20	100,000.00
20.00	5.00	1,500.00	593.60	100,000.00
20.00	10.00	1,512.00	590.60	100,000.00
20.00	15.00	1,523.00	587.30	100,000.00
20.00	20.00	1,530.00	585.80	100,000.00

Deposited Energy, E_d [MeV]	Deposited Energy, E_d [J]	Xe Density, ρ [g/cm ³]	Xe Volume [cm ³]	Xe Mass [kg]	Xe Dose [Gy]
146,888.17	2.35E-08	3.06	1,256.64	3.85	6.12E-09
149,959.96	2.40E-08	3.06	6,283.19	19.23	1.25E-09
151,223.91	2.42E-08	3.06	12,566.37	38.45	6.30E-10
152,293.71	2.44E-08	3.06	18,849.56	57.68	4.23E-10
153,018,299.59	2.45E-05	3.06	25,132.74	76.91	3.19E-07

Figure 13: Energy deposited in Xe, 2 MeV e^- beam, no Pb lid.[1][2][3]

[Activity 9] Histograms were plotted to relate different bins of energy levels to the number of energy deposition events. The bins can be as many as the sqrt(population) or at least 10 bins, refer to the project script csv2root.C, variable: nbins. This activity was performed for all five (5) instances listed in Figure 13 and each corresponding histogram is presented in Appendix C.

4.4. Simulation of a 2 MeV e^- beam, with Lead (Pb) Target

[Activity 10] A typical tracker (Xe) – target (Pb) – beam system as shown in Figure 4 was implemented, with:

trackerRadius = 20 [cm]

trackerLength = 20 [cm]

targetRadius = 20 [cm]

targetLength = [0.2, 0.4, 0.6, 0.8, 1.0] [cm]

[Activity 11] 100,000 iterations were simulated per targetLength instance, 500,000 data points in total, resulting in the chart seen in Figure 14 with tolerances as provided in Figure 15 relating Energy from a 2 MeV e^- beam, E , deposited in the tracker, as a function of the tracker thickness also known as the trackerLength: $E = 37.45 t^2 + 77.79 t + 54.05$.

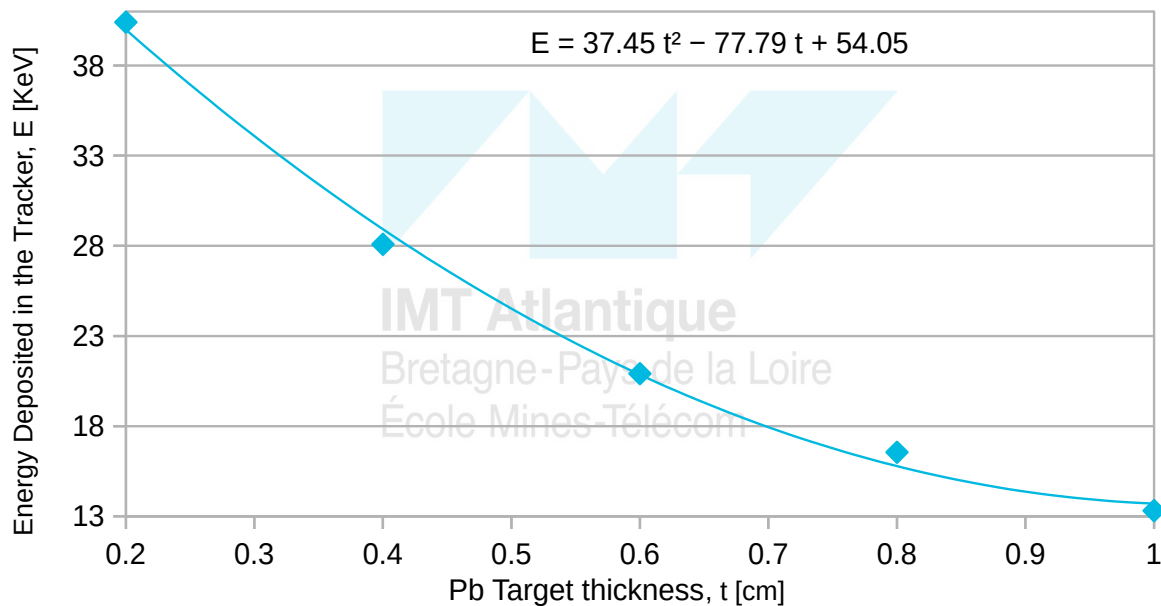


Figure 14: Energy deposited in Xe, 1 MeV, e^- beams, no Pb target.

Xe & Pb Target Radius [cm]	Pb Target Thickness [cm]	Energy deposited [keV] (Population mean, μ)	$\pm\sigma$	Population
20.00	0.20	40.40	174.20	100,000.00
20.00	0.40	28.08	151.70	100,000.00
20.00	0.60	20.91	134.40	100,000.00
20.00	0.80	16.55	121.80	100,000.00
20.00	1.00	13.31	110.60	100,000.00

Deposited Energy, E_d [MeV]	Deposited Energy, E_d [J]	Xe Density, ρ [g/cm ³]	Xe Volume [cm ³]	Xe Mass [kg]	Xe Dose [Gy]
4,039.88	6.47E-10	3.06	25,132.74	76.91	8.42E-12
2,808.48	4.50E-10	3.06	25,132.74	76.91	5.85E-12
2,091.18	3.35E-10	3.06	25,132.74	76.91	4.36E-12
1,654.68	2.65E-10	3.06	25,132.74	76.91	3.45E-12
1,331.25	2.13E-10	3.06	25,132.74	76.91	2.77E-12

Figure 15: Energy deposited in Xe, 2 MeV e^- beam, with Pb target.[1][2][3]

[Activity 12] Histograms were plotted to relate different bins of energy levels to the number of energy deposition events. The bins can be as many as the $\sqrt{\text{population}}$ or at least 10 bins, refer to the project script `csv2root.C`, variable: `nbins`. This activity was performed for all five (5) instances listed in Figure 15 and each corresponding histogram is presented in Appendix D.

5. Conclusions

[Conclusion 1] It was observed that the use of lead (Pb) sheets effectively shielded the Xenon energy trackers from the effects of radiation energy deposition (E), from which the best performing were geometries with a thickness (t) of 1 [cm], with the following characteristic curves:

As seen in Figure 10, for a 1 MeV γ radiation beam: $E = 76.07 t^2 + 478.49 t + 919.04$.

As seen in Figure 14, for a 2 MeV e^- radiation beam: $E = 37.45 t^2 + 77.79 t + 54.05$.

[Conclusion 2] It was observed that, the more mass a given 'tracker' has as a function of its reference cylinder length (x), the more energy it will absorb according to the following characteristic curves (E), with the following characteristic curves:

As seen in Figure 8, for a 1 MeV γ radiation beam: $E = -2.31 x^2 + 91.21 x + 8.35$.

As seen in Figure 12, for a 2 MeV e^- radiation beam: $E = -0.17 x^2 + 6.54 x + 1465.60$.

[Conclusion 3] The analysis of the histograms in Appendix A, Appendix B, Appendix C, and Appendix D, reveals that the tracker is shielded from the energy deposition effects of both γ particle and e^- particle radiation by implementing the 0.2 through 1 cm shields. However, a thicker shield may be needed to effectively prevent the radiation energy deposition in the tracker if at all.

6. References

- [1] Bureau International des Poids et Mesures (2024). The International System of Units (SI). 9th ed. [online] Bureau International des Poids et Mesures. Available at: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9.pdf> [Accessed 31 Jan. 2026].
- [2] DAVESNE, E. (2025). Health Effects Of Ionizing Radiations. Reglementation-control.easnr.fr: ASNR, p.14. Master Class at IMT Atlantique, 14 April 2026.
- [3] SIHVER, L. (2026). Short Repetition of Dosimetric Units & Radiation Quantities. UTRGV.edu: UTRGV, p.50. Dosimetric Units & Quantities for Radiation Protection. 2026 Master Class at IMT Atlantique.

Appendix A. 1 MeV γ -rays, No Pb Target. Histograms.

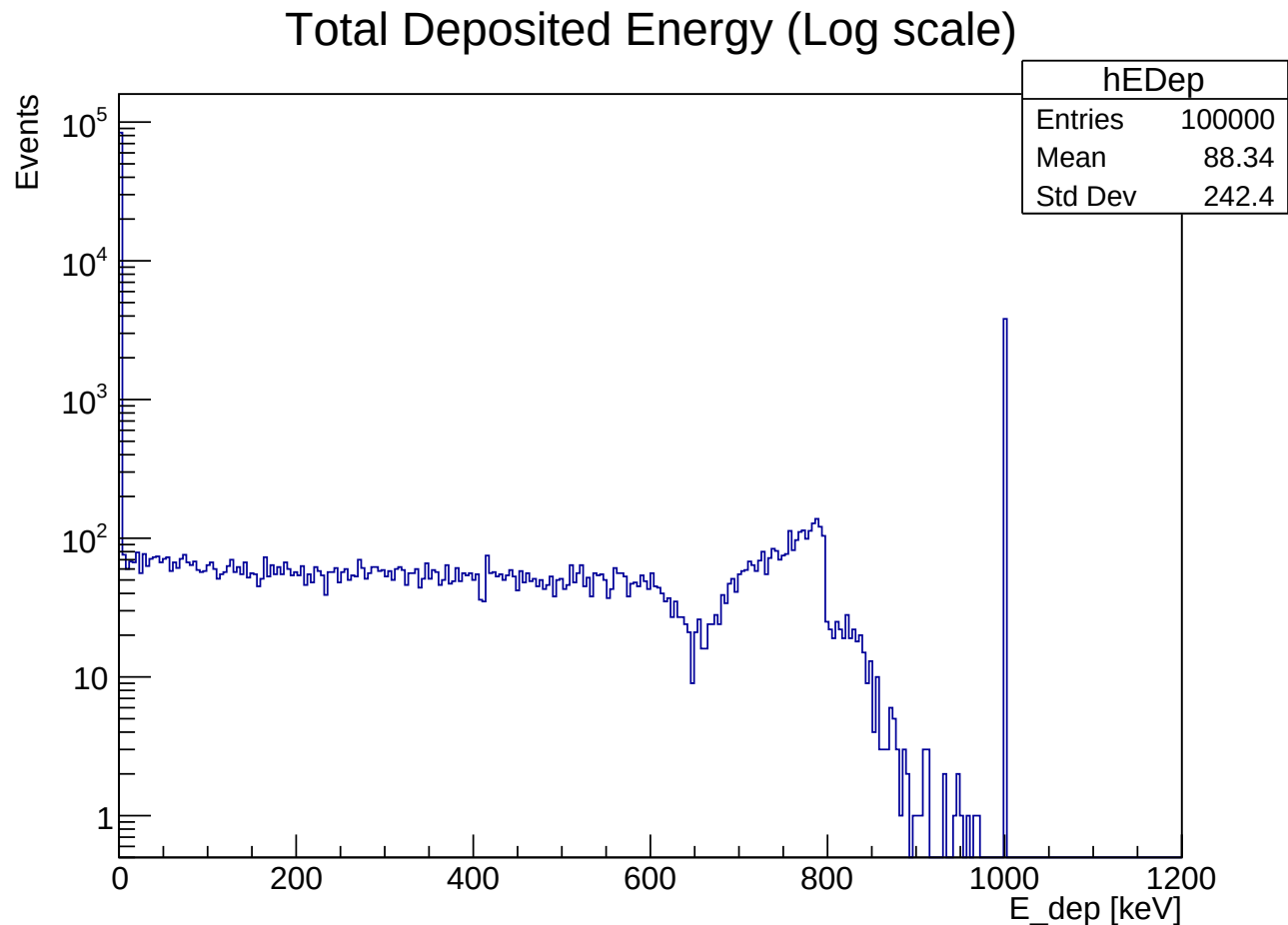


Figure 16: 1cm-1MeV-100,000.00 γ -no Pb

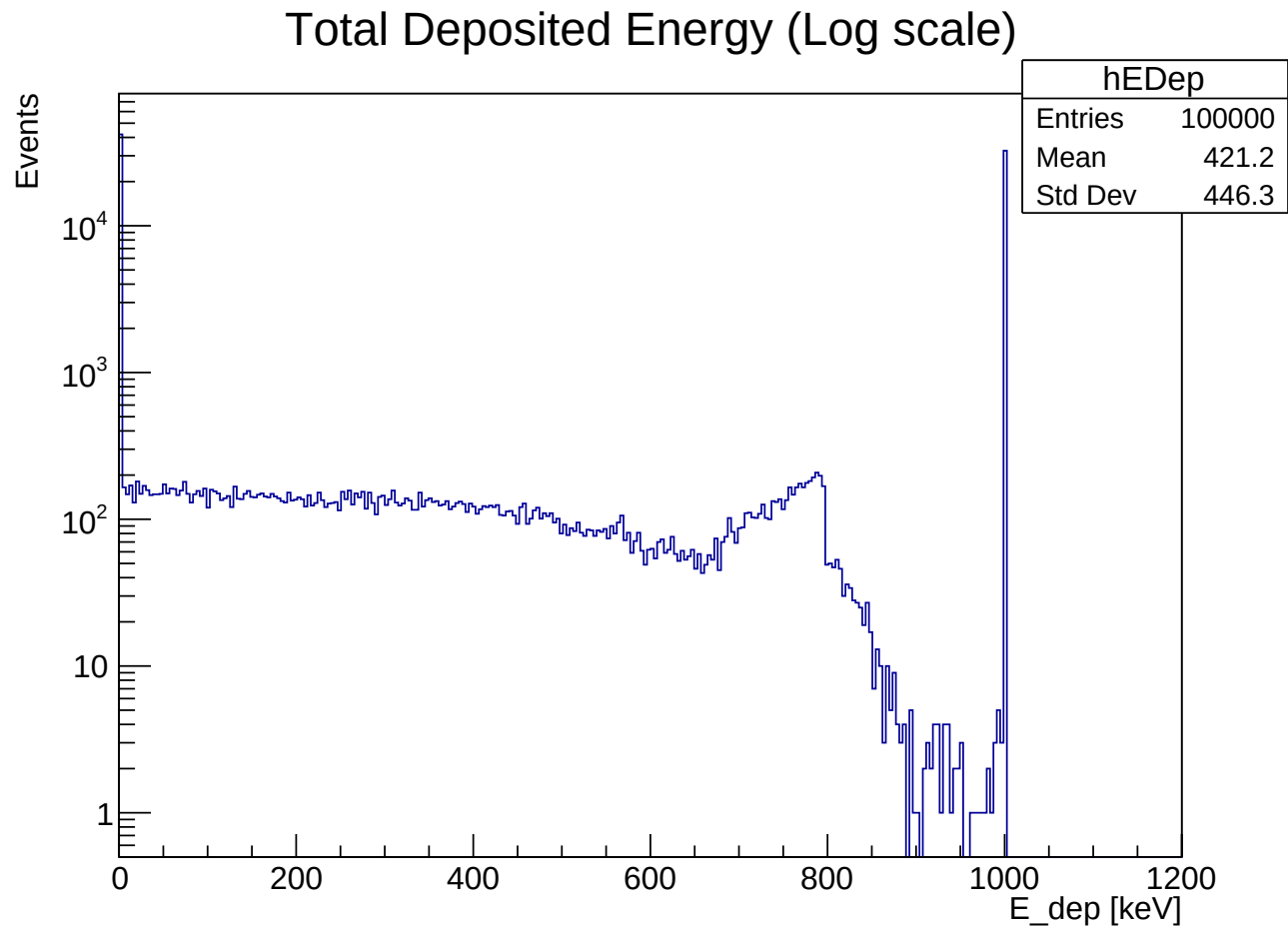


Figure 17: 5cm-1MeV-100,000.00γ-no Pb

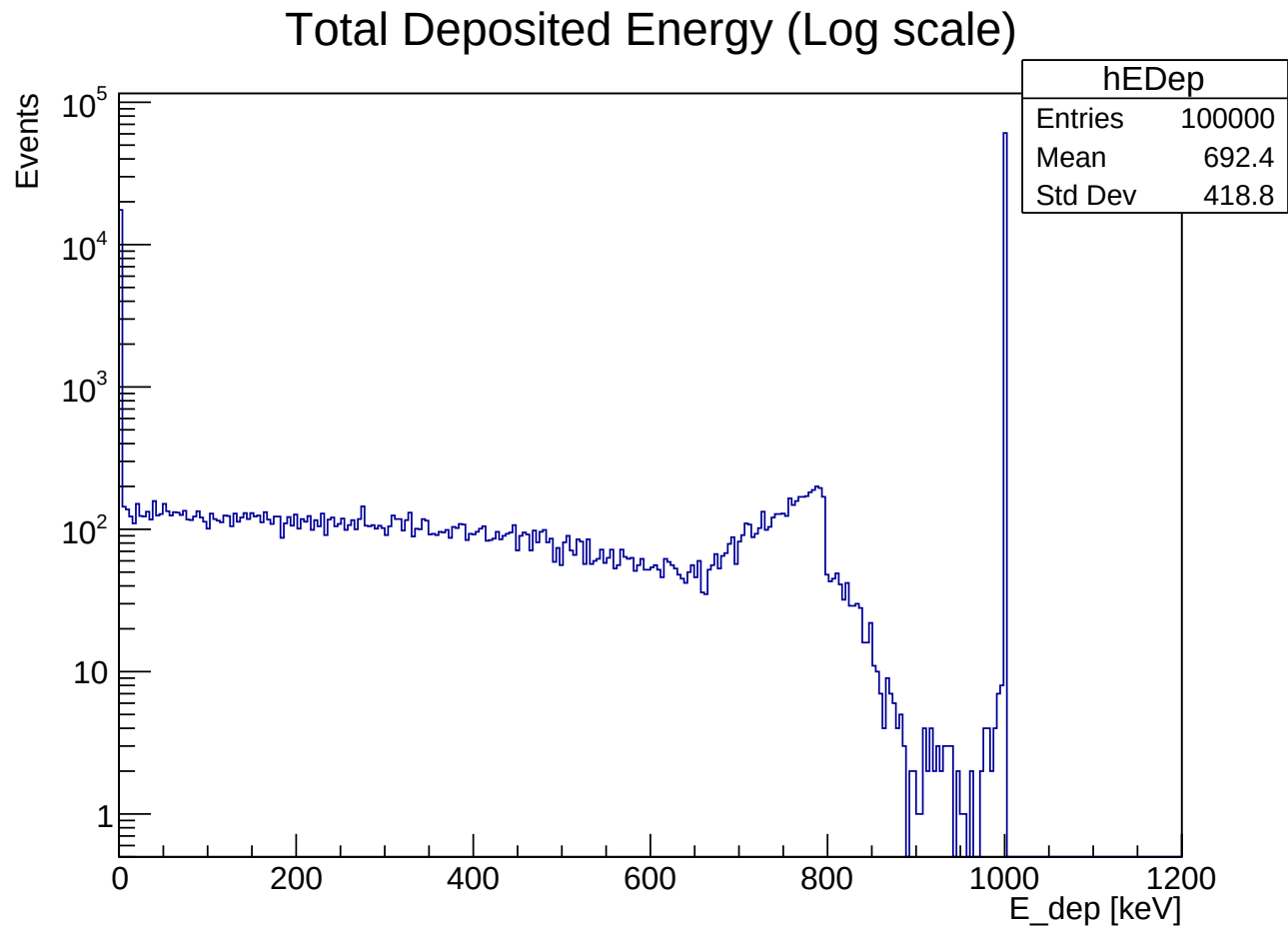


Figure 18: 10cm-1MeV-100,000.00γ-no Pb

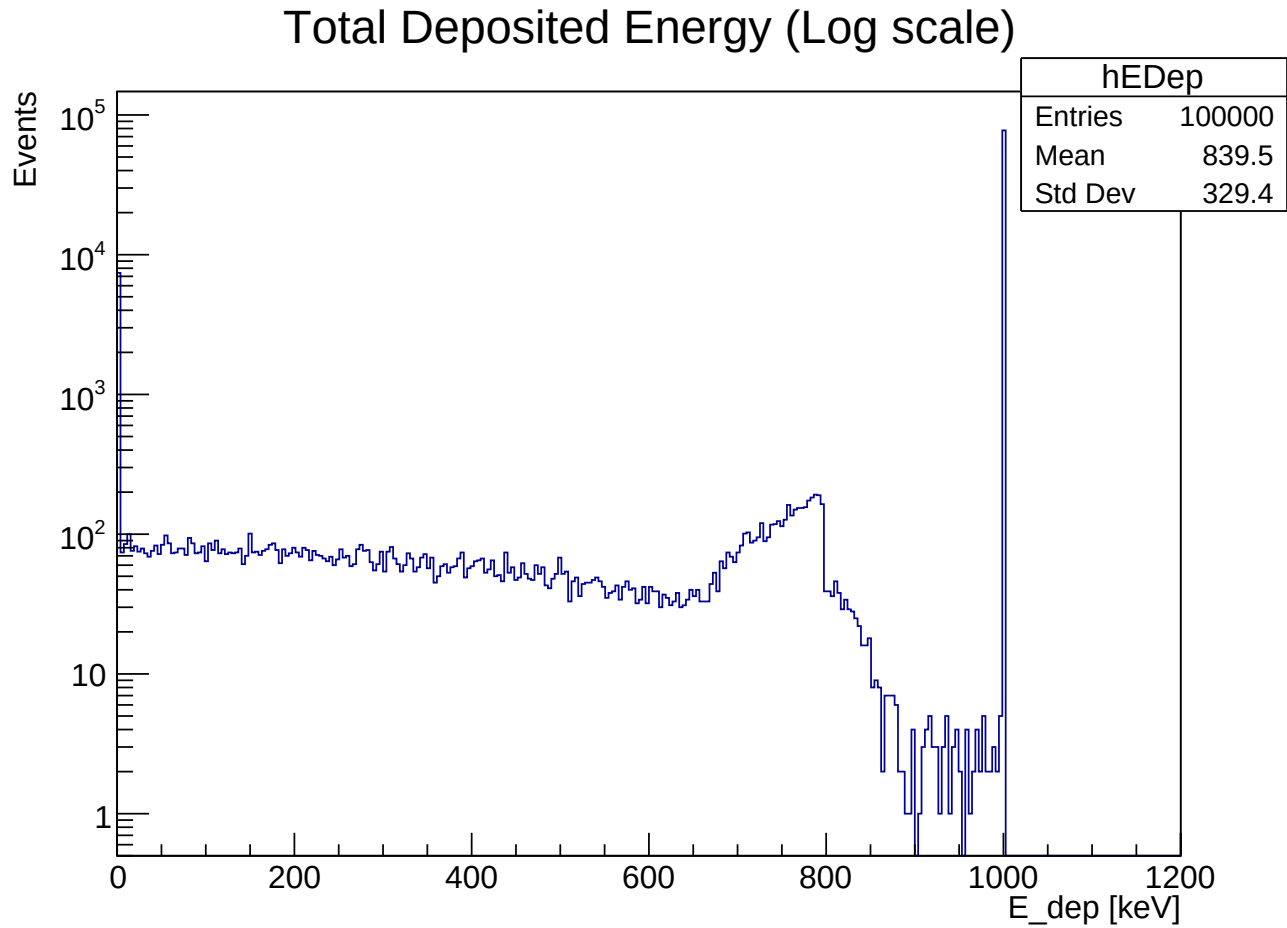


Figure 19: 15cm-1MeV-100,000.00γ-no Pb

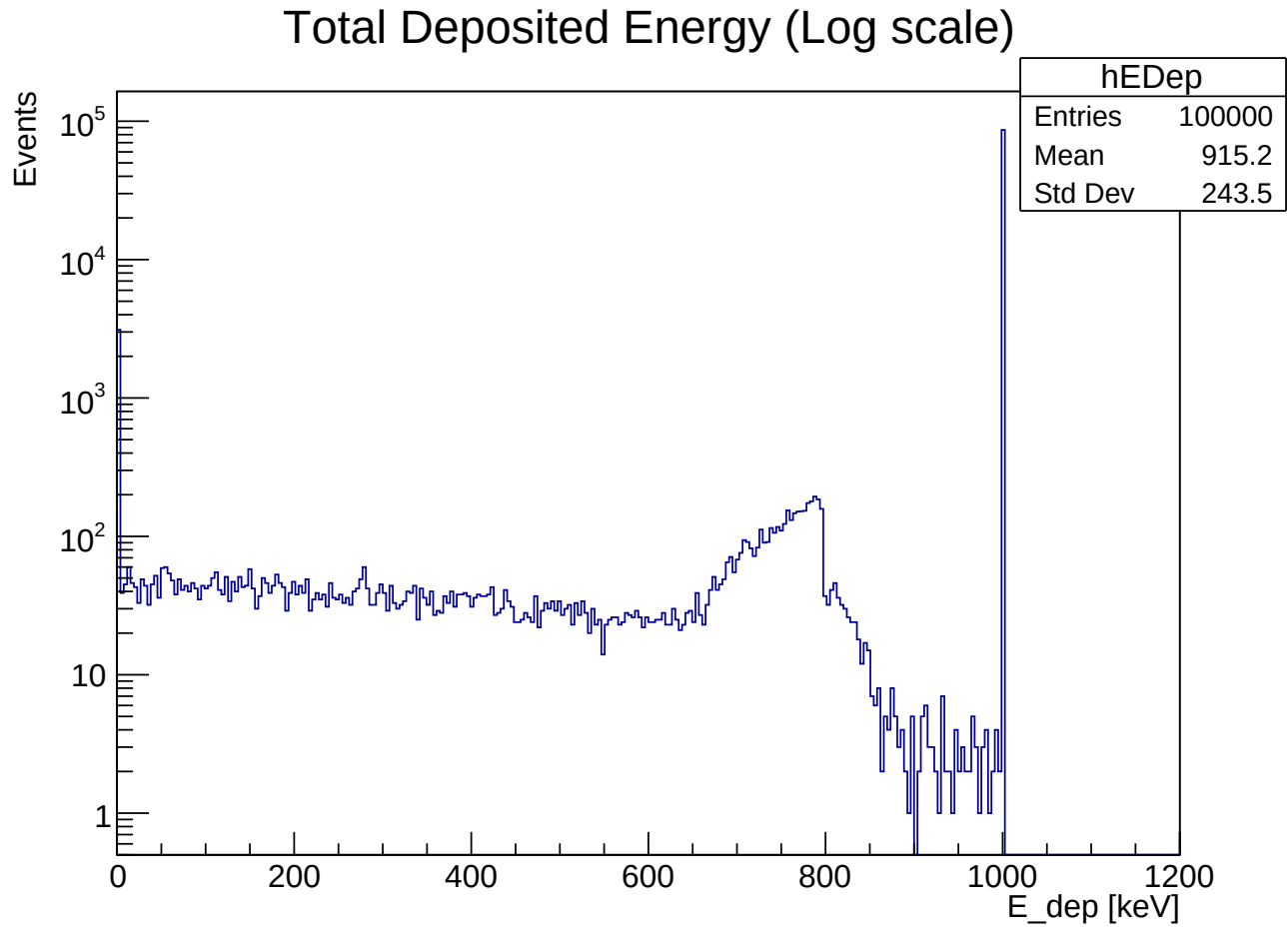


Figure 20: 20cm-1MeV-100,000.00γ-no Pb

Appendix B. 1 MeV γ -rays, with Pb Target. Histograms.

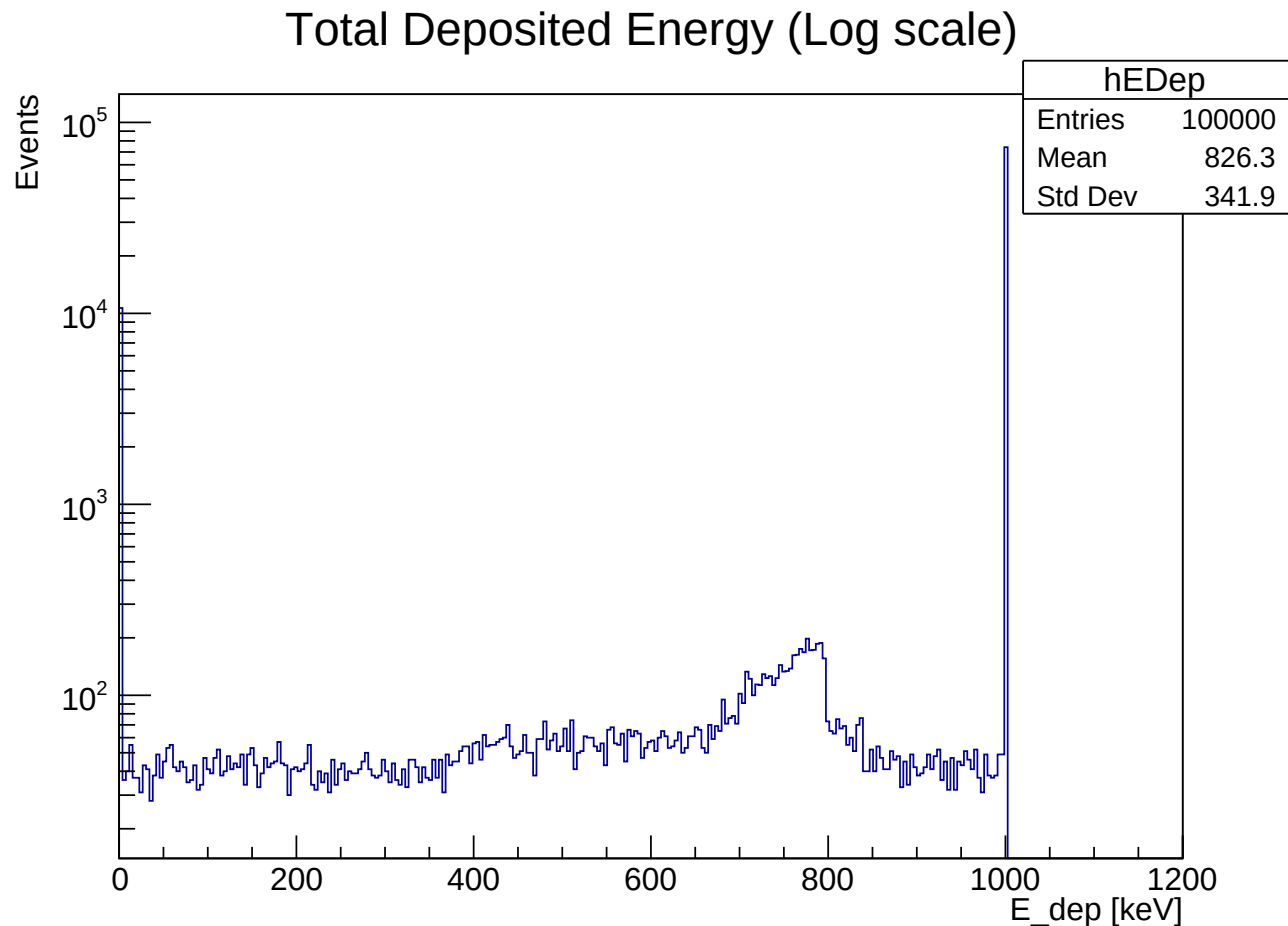


Figure 21: 0.2 cm-1MeV-100,000.00 γ -with Pb

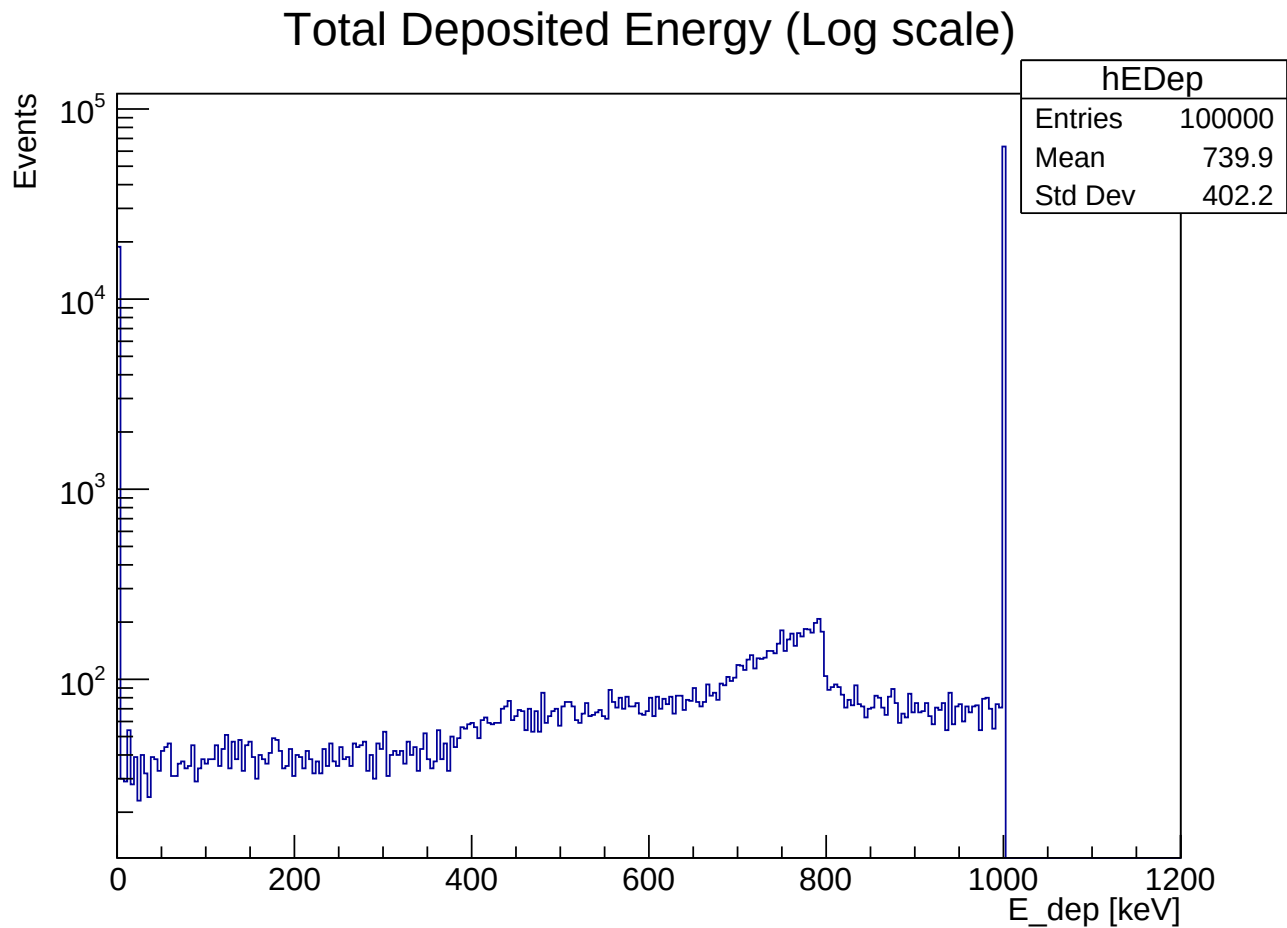


Figure 22: 0.4cm-1MeV-100,000.00γ-with Pb

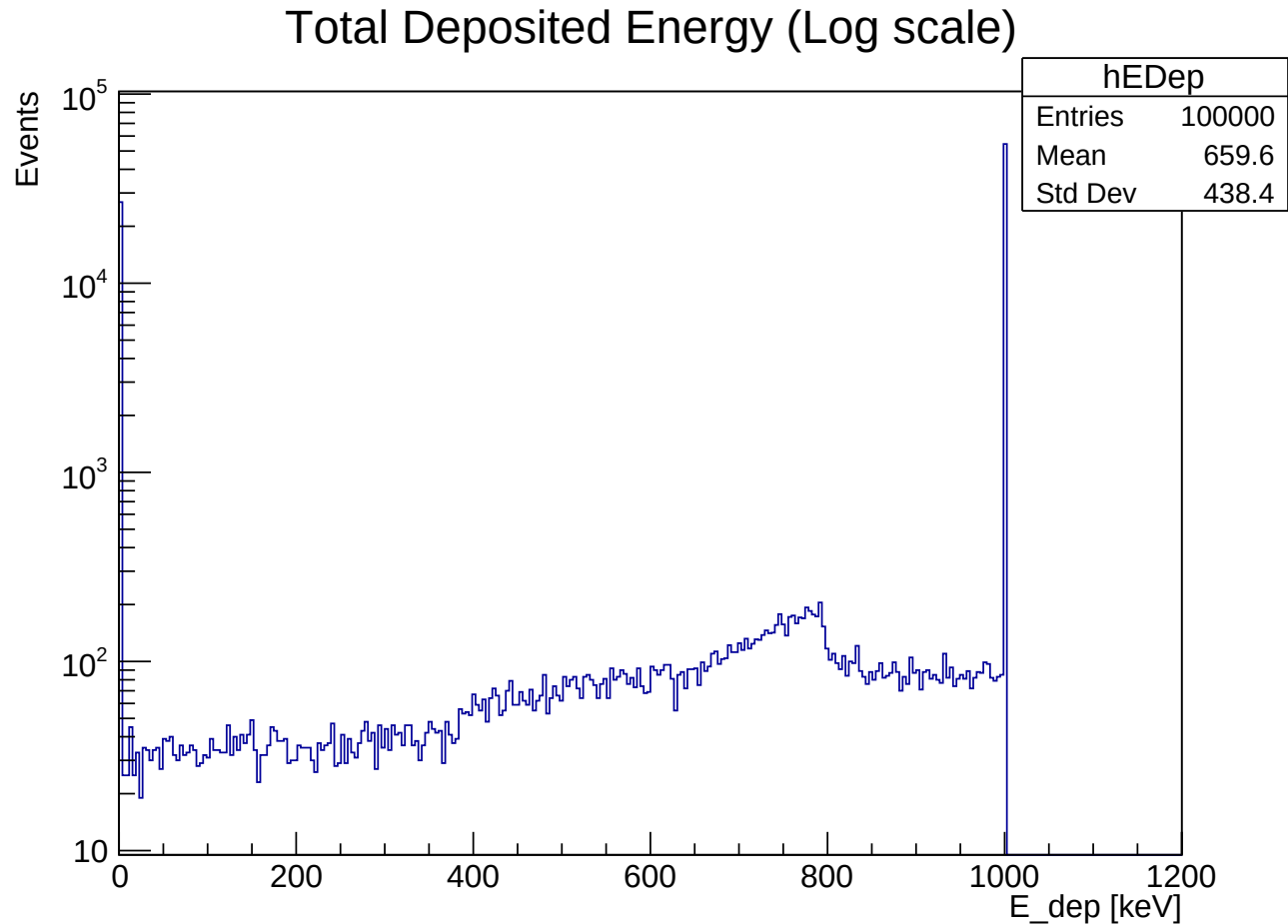


Figure 23: 0.6cm-1MeV-100,000.00y-with Pb

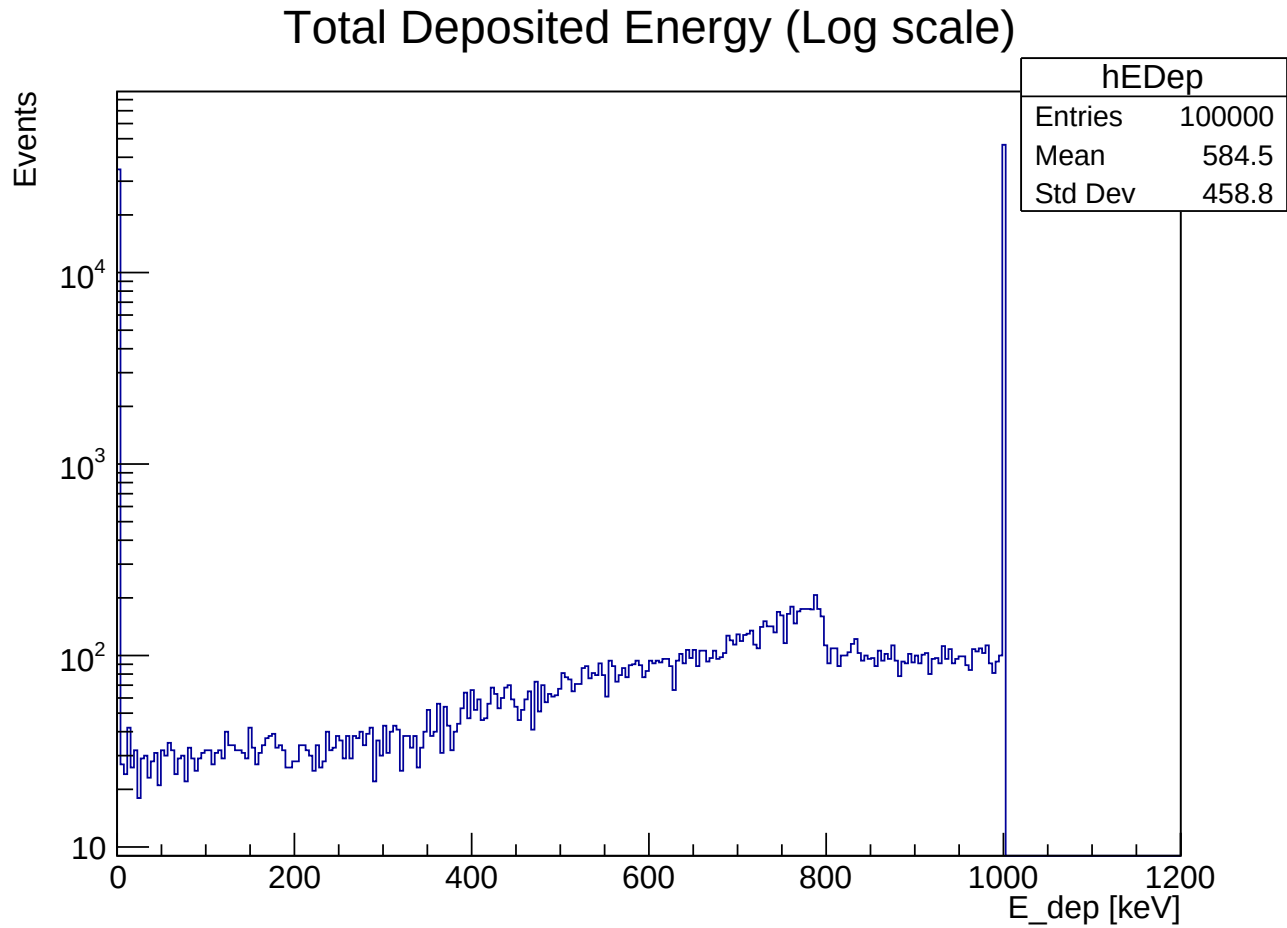


Figure 24: 0.8cm-1MeV-100,000.00γ-with Pb

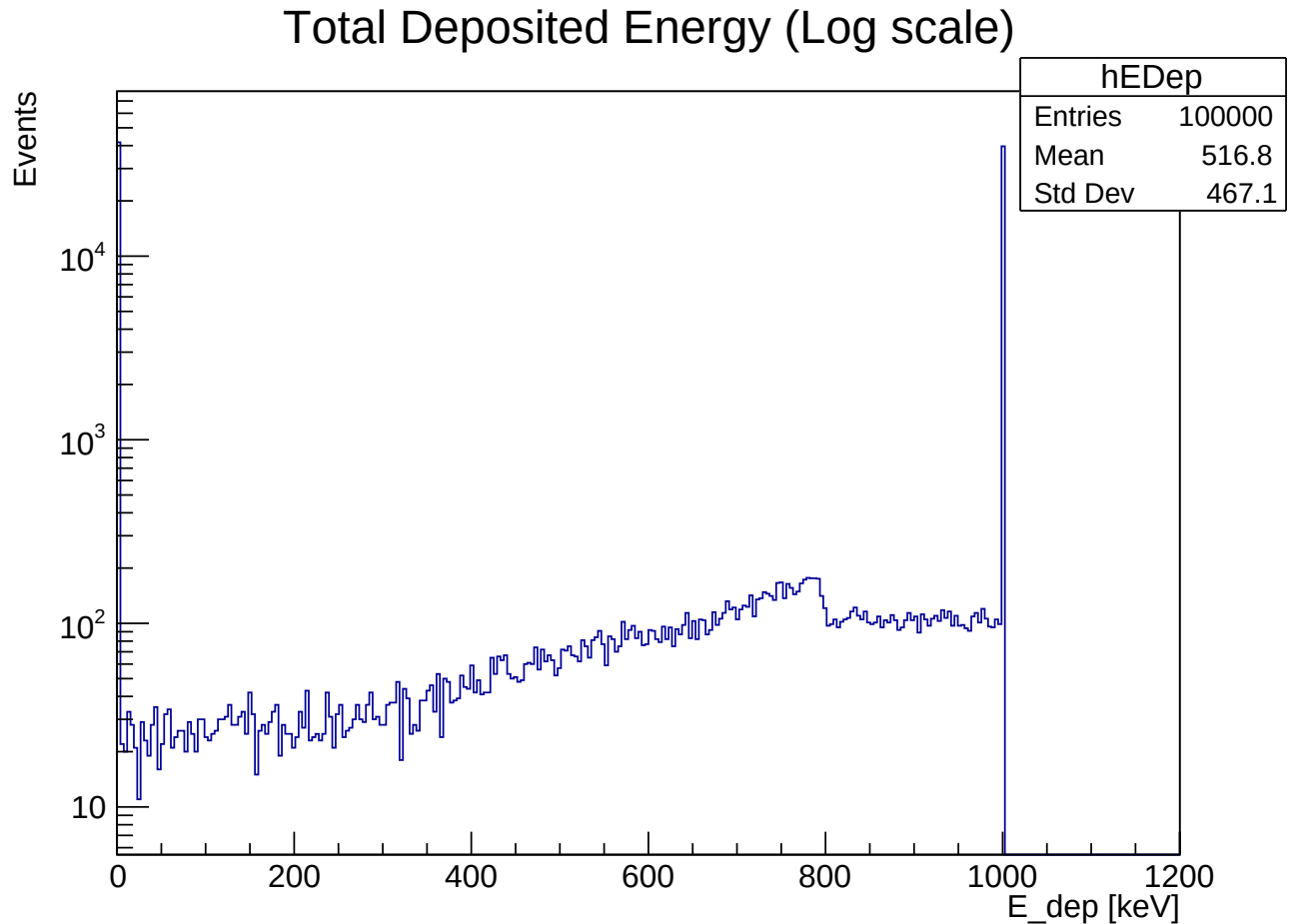


Figure 25: 1.0cm-1MeV-100,000.00γ- with Pb

Appendix C. 2 MeV e^- -beams, No Pb Target. Histograms.

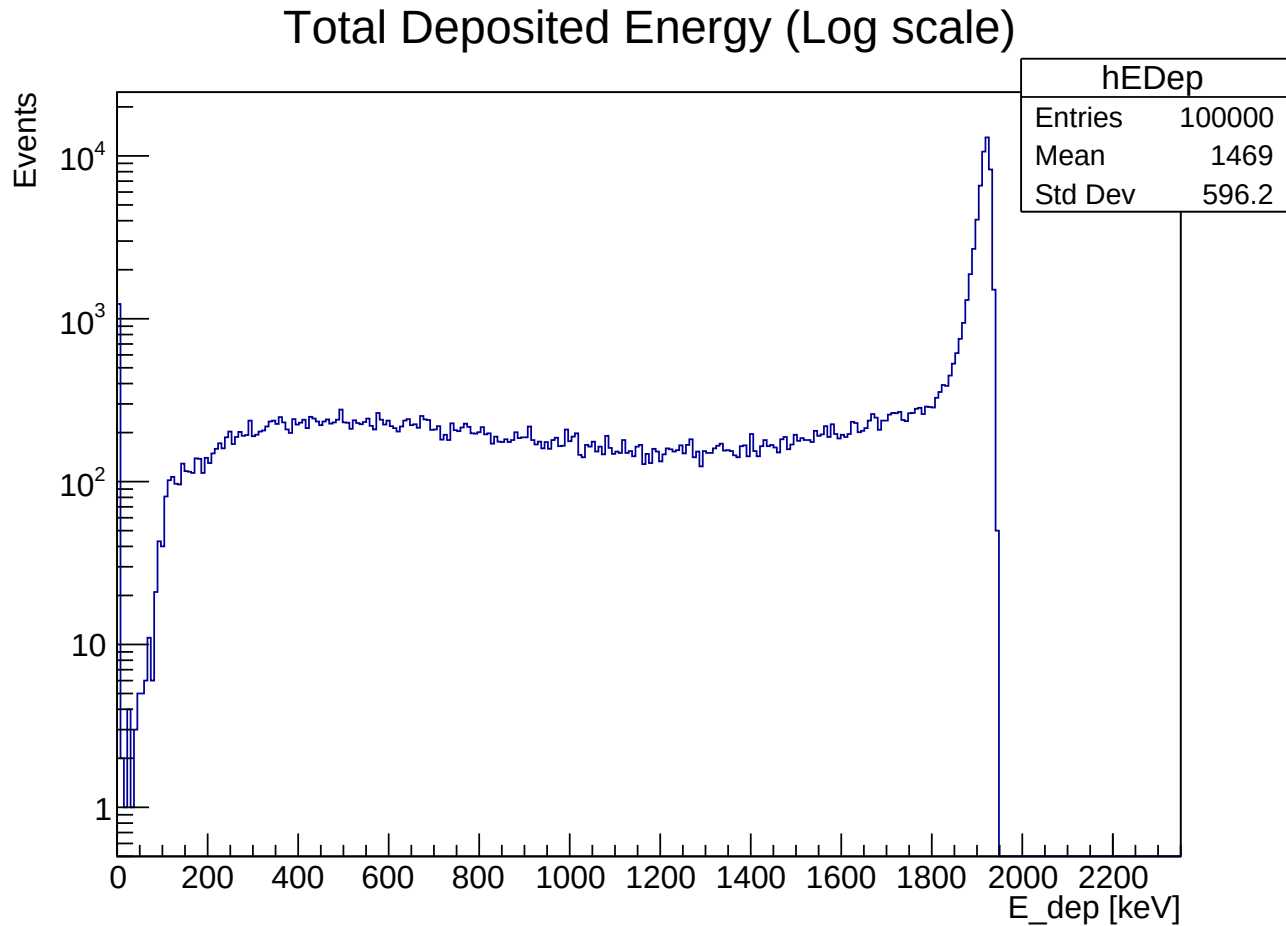


Figure 26: 1cm-2MeV-100,000.00e⁻-no Pb

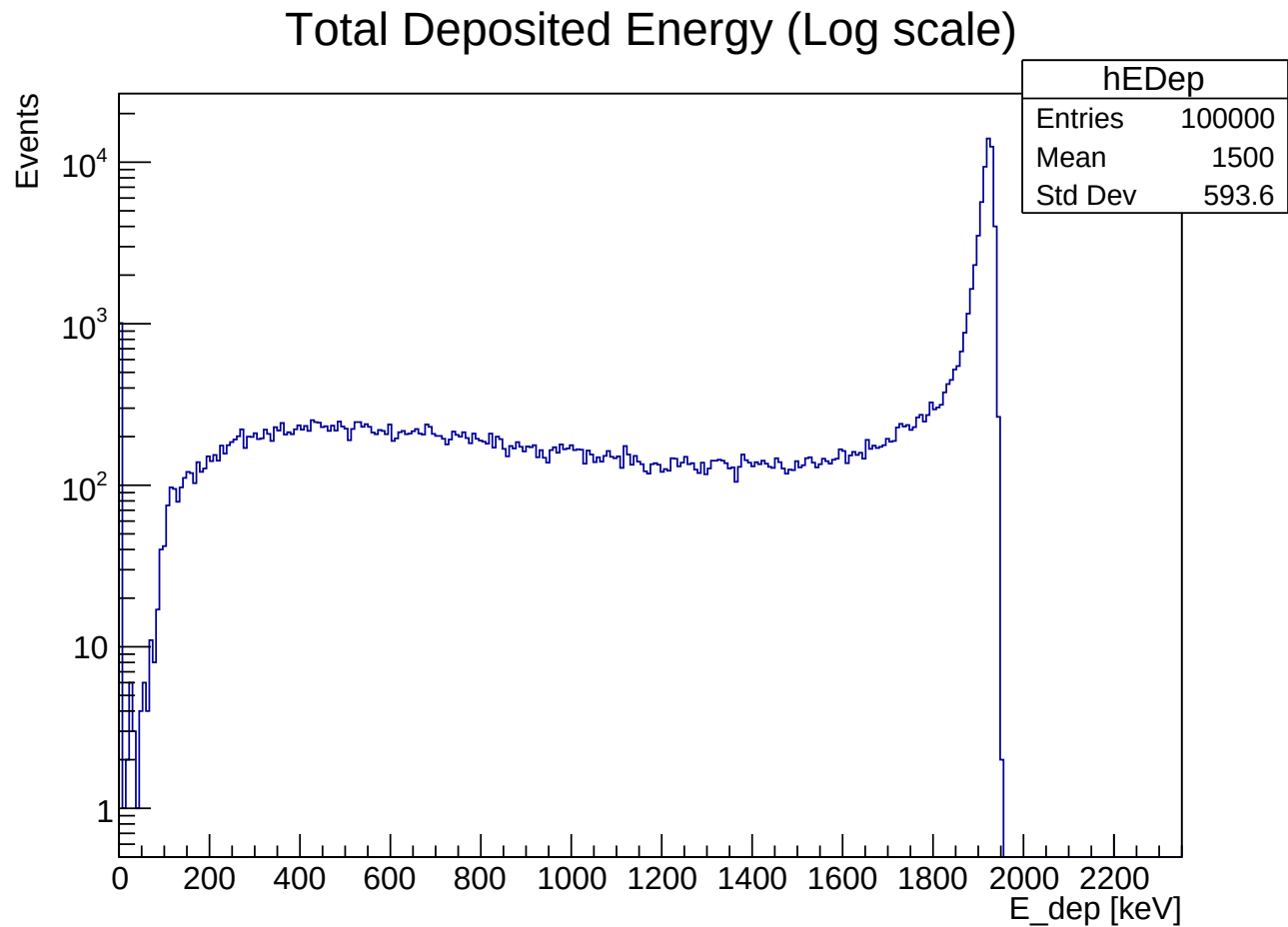


Figure 27: 5cm-2MeV-100,000.00e⁻-no Pb

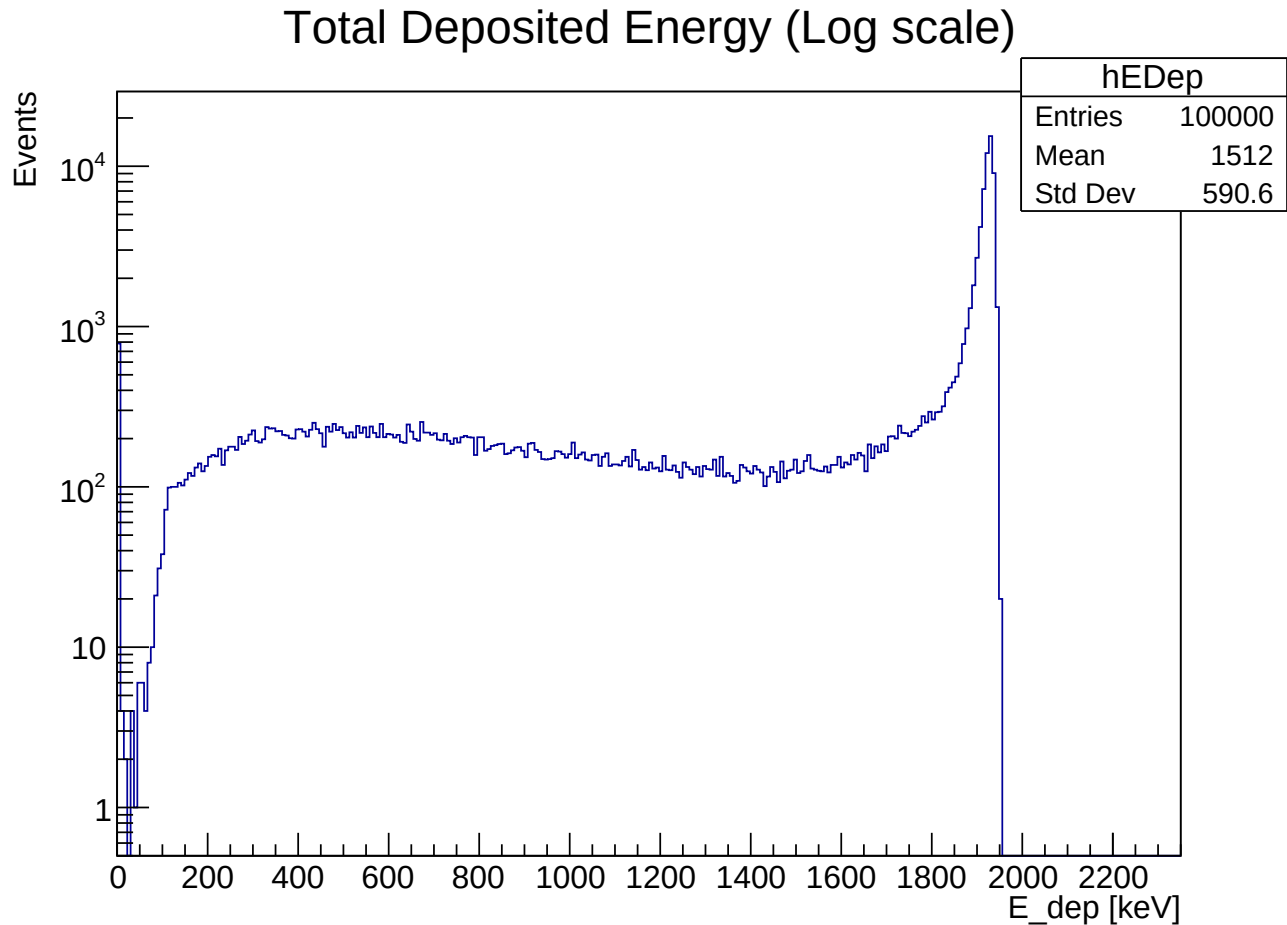


Figure 28: 10cm-2MeV-100,000.00e⁻-no Pb

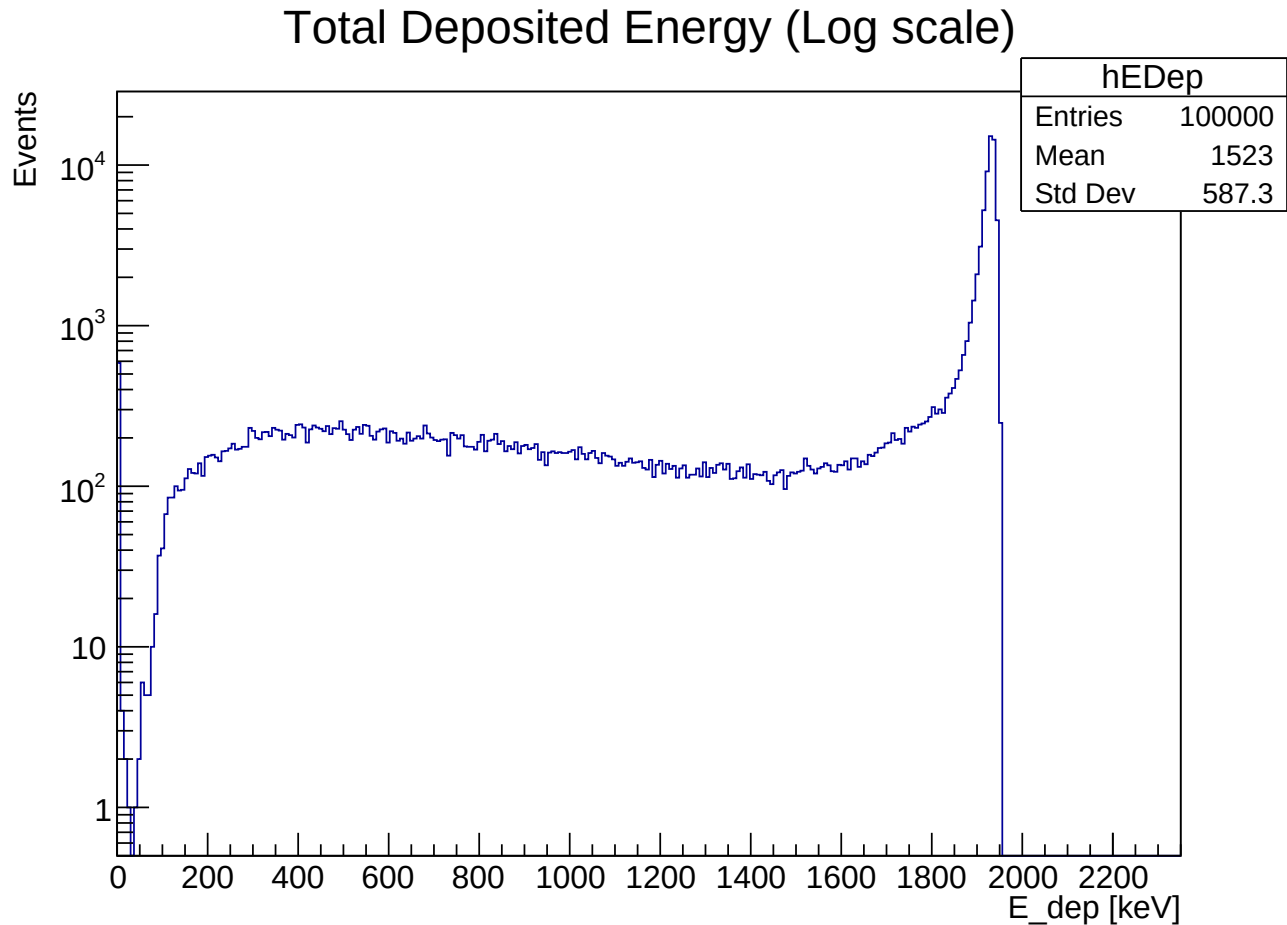


Figure 29: 15cm-2MeV-100,000.00e⁻-no Pb

Total Deposited Energy (Log scale)

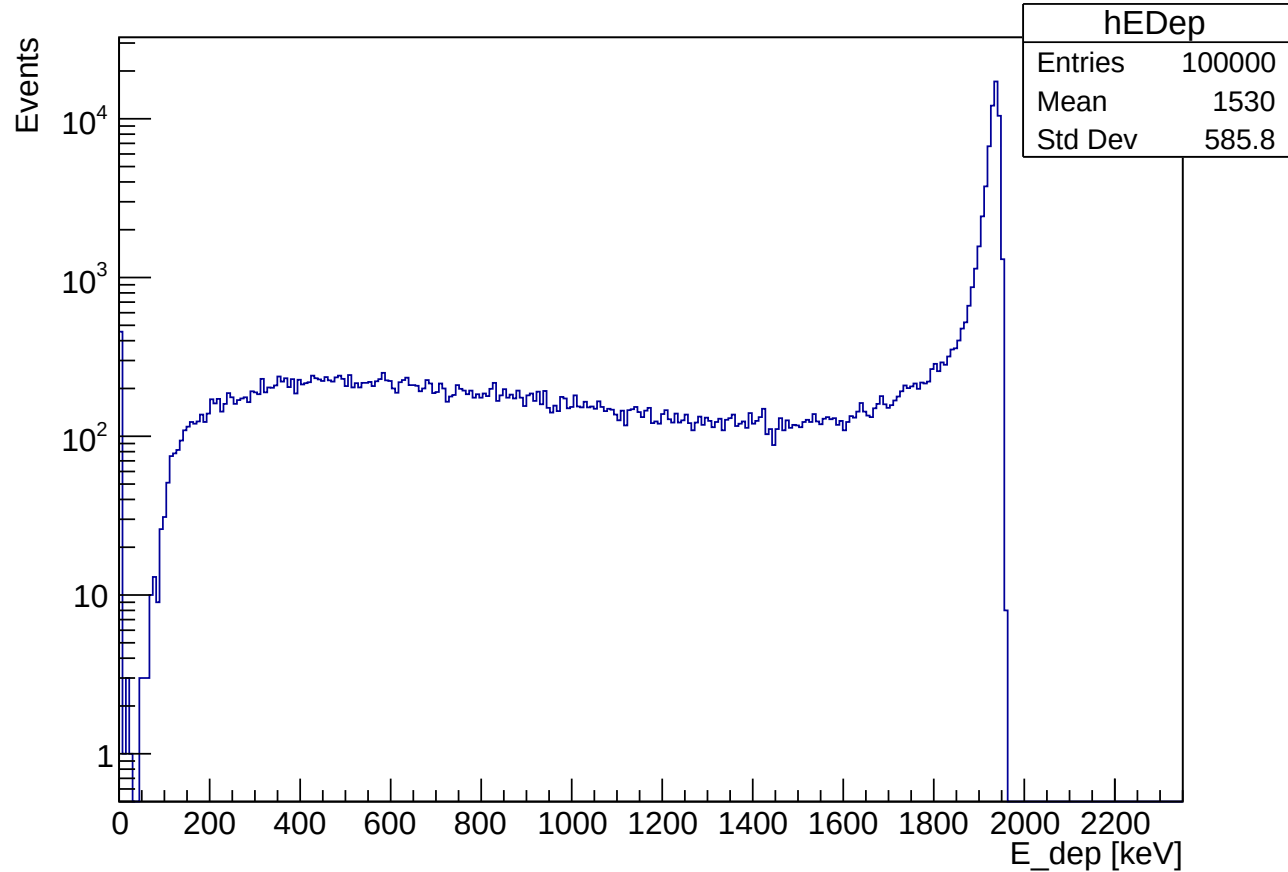


Figure 30: 20cm-2MeV-100,000.00e⁻-no Pb

Appendix D. 2 MeV e^- -beams, with Pb Target. Histograms.

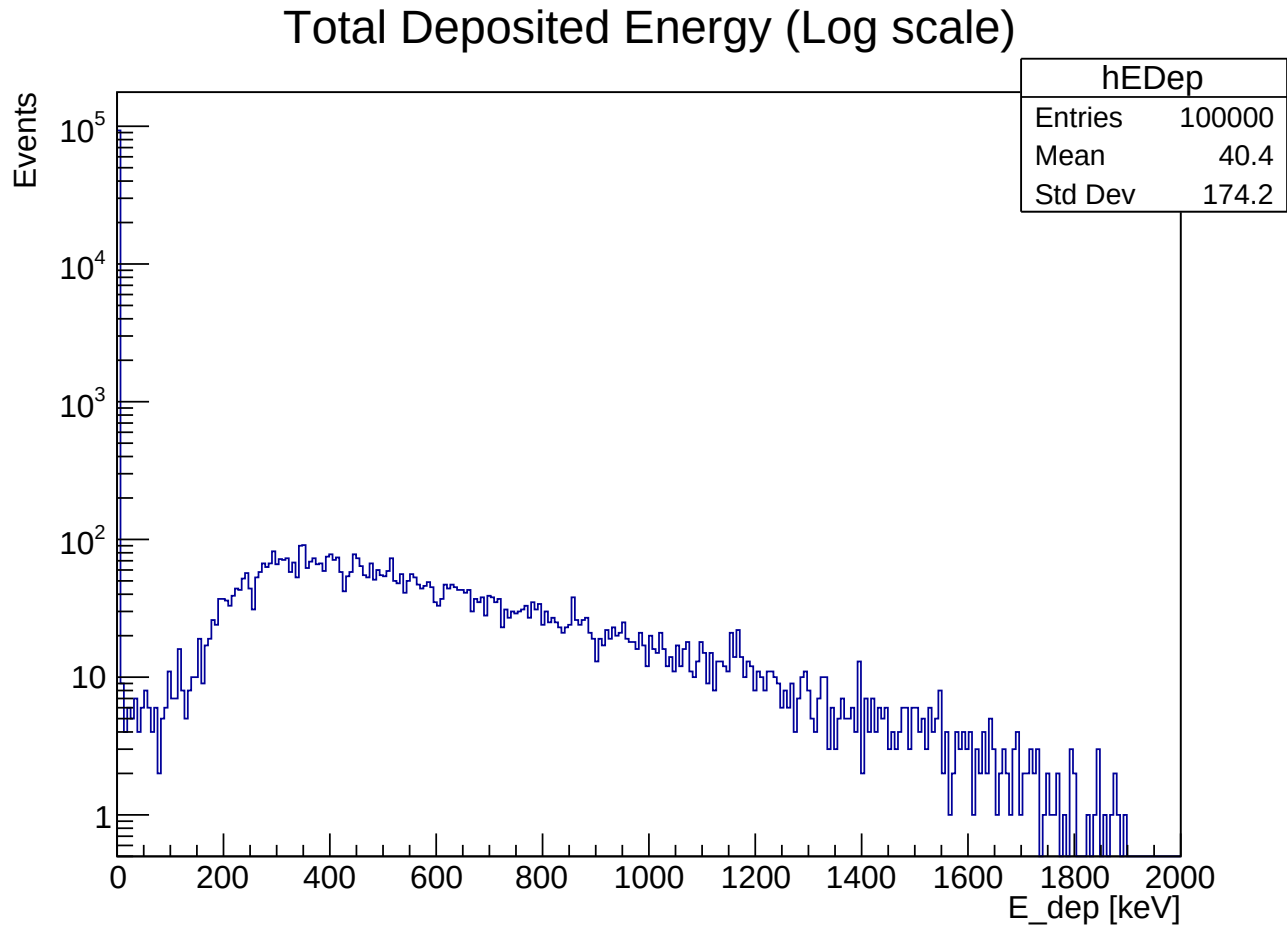


Figure 31: 0.2cm-2MeV-100,000.00 e^- -with Pb

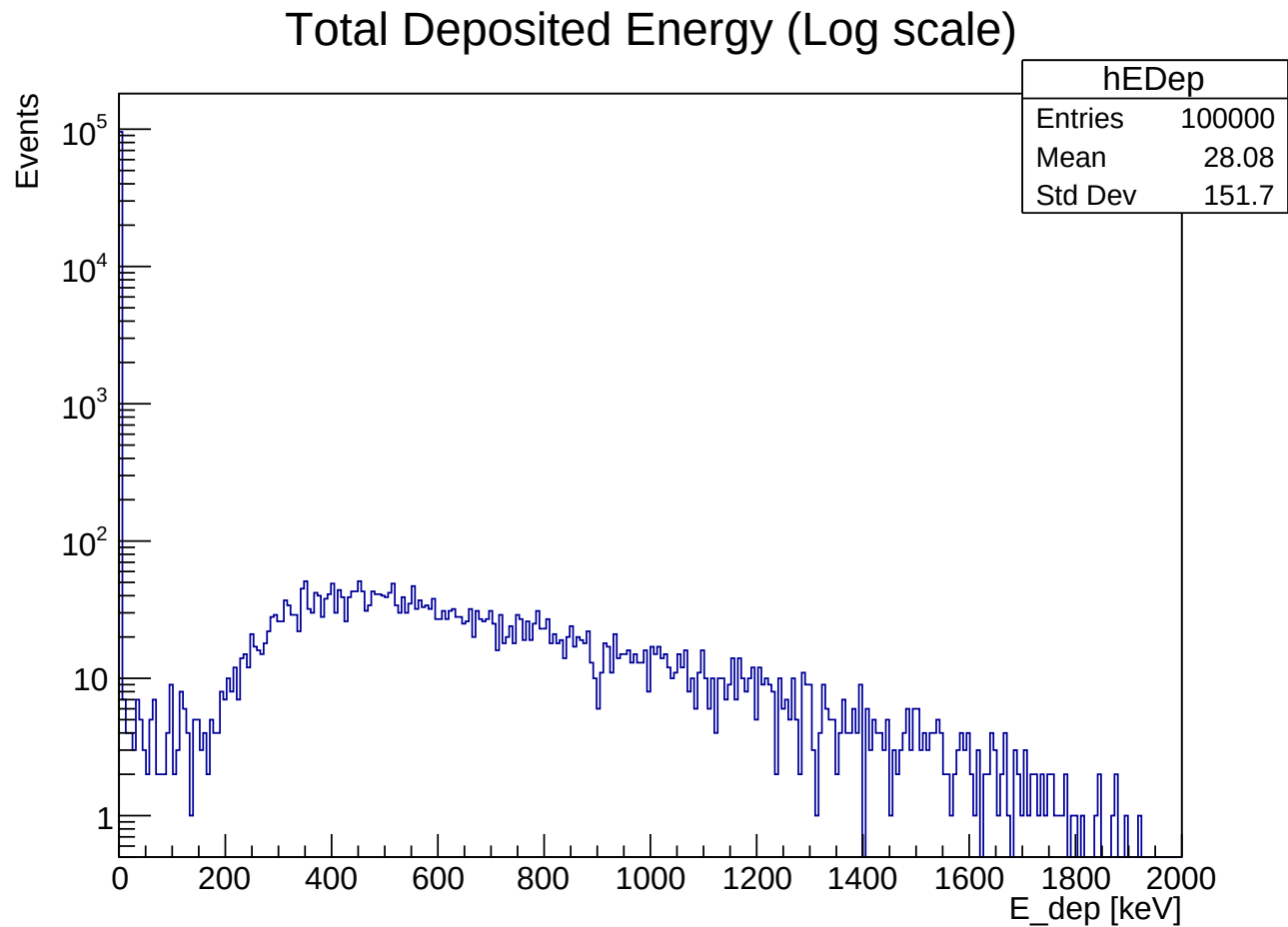


Figure 32: 0.4cm-2MeV-100,000.00e⁻-with Pb

Total Deposited Energy (Log scale)

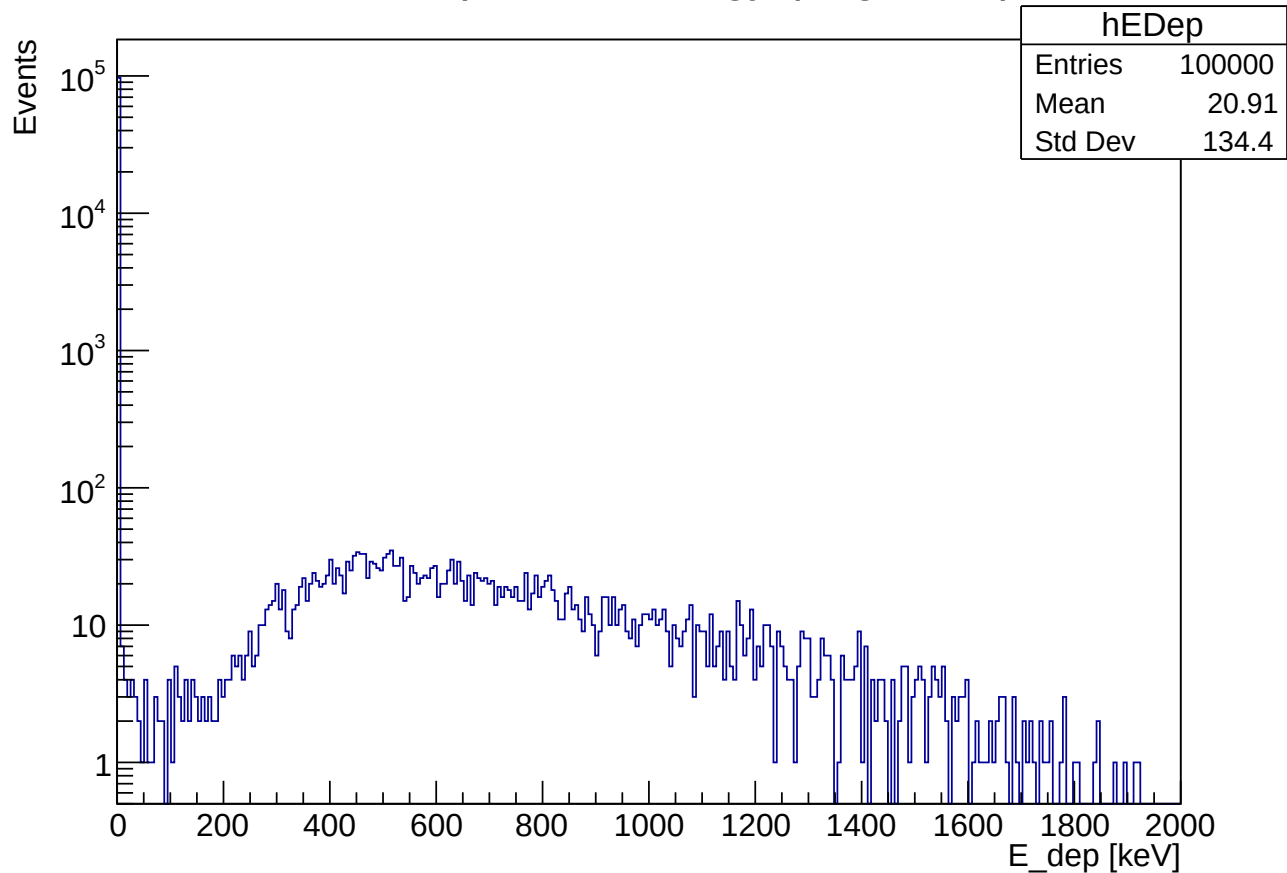


Figure 33: 0.6cm-2MeV-100,000.00e⁻-with Pb

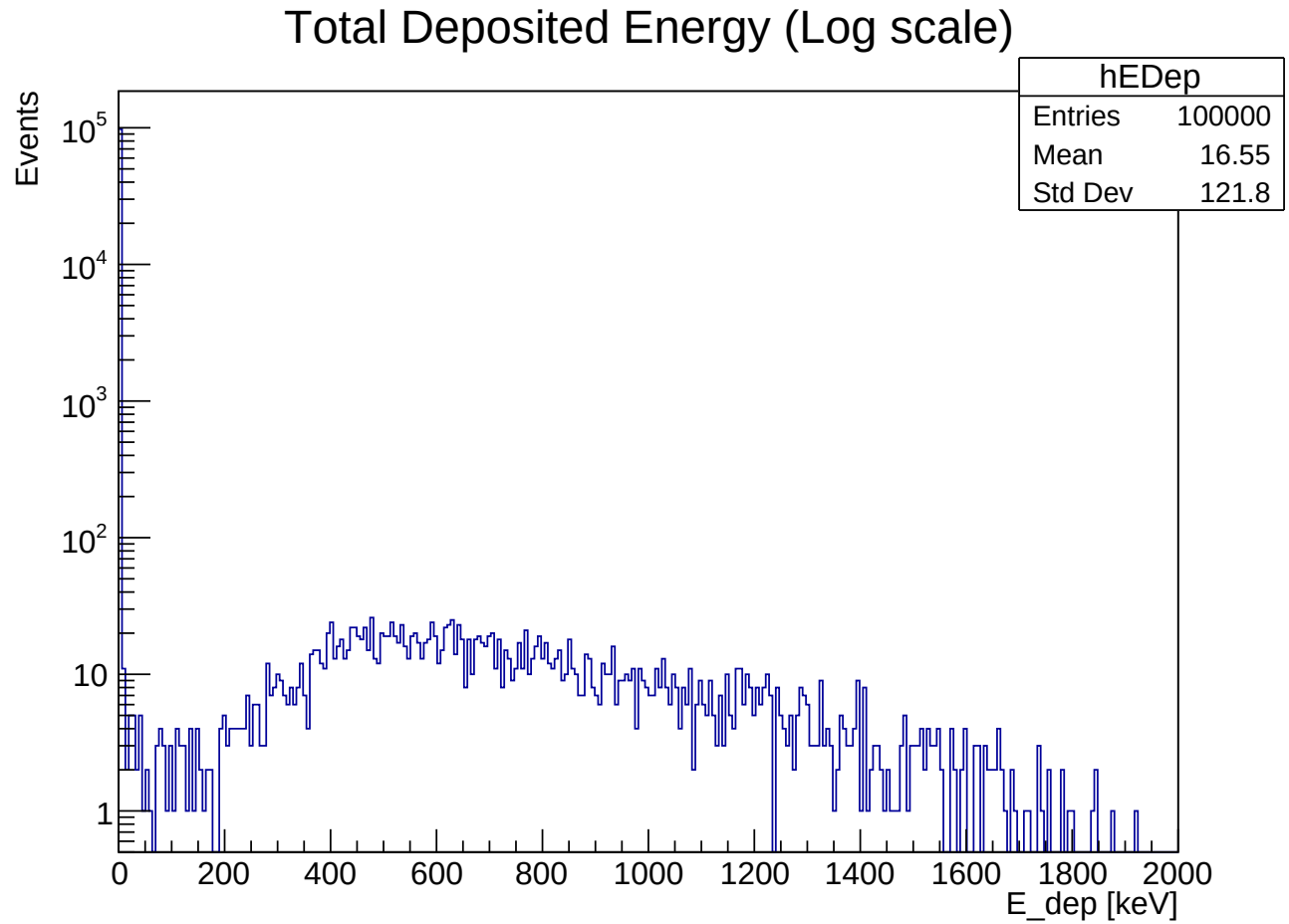


Figure 34: 0.8cm-2MeV-100,000.00e⁻-with Pb

Total Deposited Energy (Log scale)

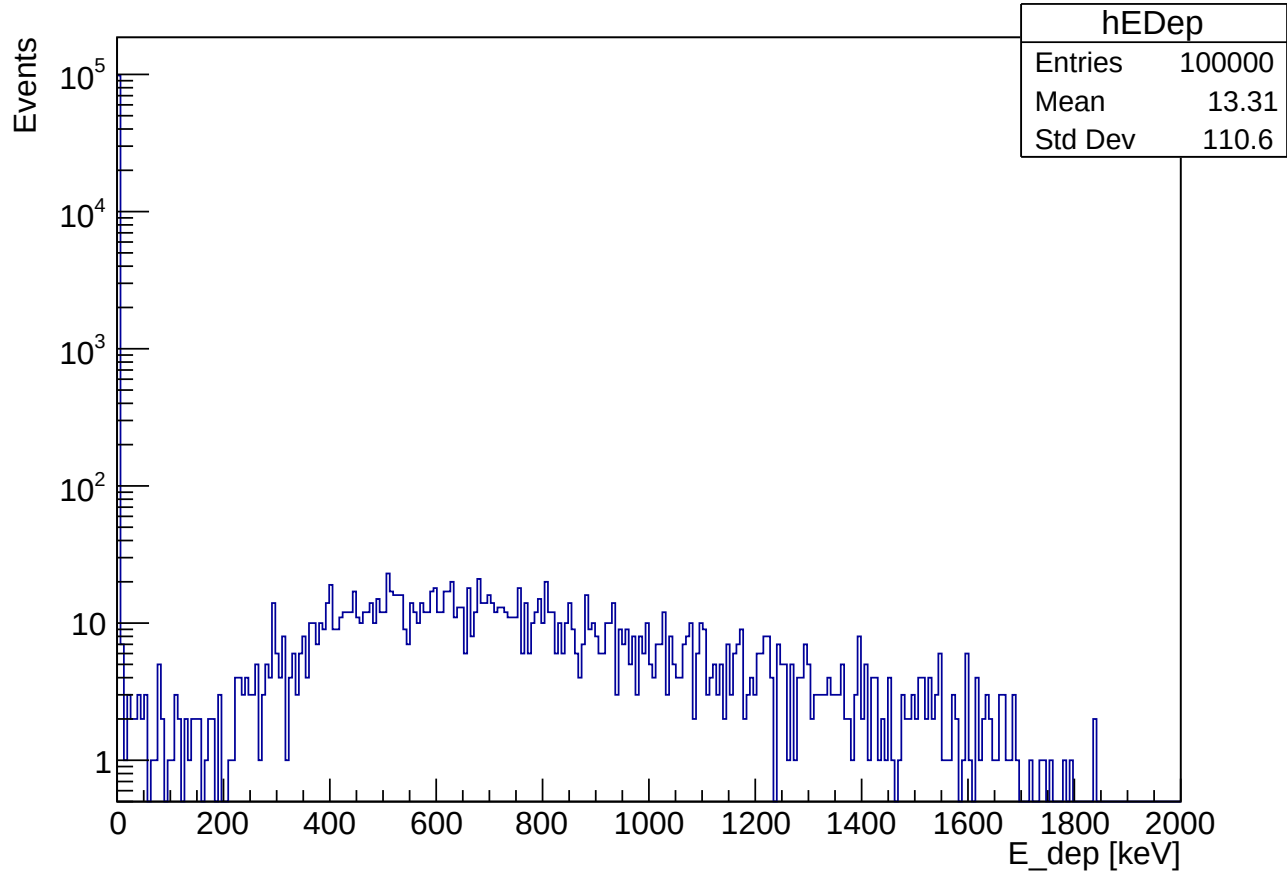


Figure 35: 1.0cm-2MeV-100,000.00e⁻-with Pb